

Chapter 55 — Management of Abnormalities of the External Genitalia in Males

- The normal penis comprises the **prepuce (foreskin) and frenulum, glans, urethral meatus, coronal sulcus, and a shaft of two corpora cavernosa with the urethra**. The ventral median raphe is continuous with the **scrotal raphe** and ends at the edge of the outer prepuce.
- Penile and scrotal anomalies are **common** and may be **congenital, acquired or iatrogenic**. Congenital anomalies arise from disorders of **sexual differentiation, genital differentiation or genital growth**, and may be part of a syndrome or involve other organ systems — **up to half of children with anorectal malformations have an associated urological malformation**.
- **Fetal sex determination by ultrasound is not accurate enough before 12 weeks' gestation; after 13 weeks it is accurate in at least 99% of fetuses without malformed genitalia**.
- Antenatal ultrasound has identified **ambiguous genitalia and penoscrotal transposition**. **Three-dimensional (3D) ultrasound** is being used to examine the fetal genitalia, but there is **no evidence yet that it improves diagnosis**.
- Most neonatal genital abnormalities do **not** produce ambiguity; far commoner are **inconspicuous penis and abnormal penile orientation**. Recognising them depends on knowing normal anatomy and embryology. The American Academy of Pediatrics (AAP) published a guide (2000) to normal and altered genital embryology and to the evaluation and management of ambiguous genitalia.

NORMAL MALE EXTERNAL GENITALIA

Embryology

- The external genitalia of the two sexes are **identical until the ninth week of gestation**.
- Male differentiation requires an intact chain: **testosterone synthesis by fetal Leydig cells → conversion to dihydrotestosterone (DHT) by 5 α -reductase → androgen receptors able to recognise the hormone**.
- **DHT acting on androgen receptors during weeks 9–13** converts the undifferentiated primordia into male structures; without androgen action they **passively** become female structures.

FATE OF THE EXTERNAL GENITAL PRIMORDIA

Primordium	Male (DHT-driven, weeks 9–13)	Female / undervirilised (passive)
Genital tubercle	Glans penis	Clitoris
Genital (urethral) folds	Penile shaft — ventral folds form the lateral walls of the urethral groove and fuse in the midline	Labia minora
Genital (labioscrotal) swellings	Scrotum — inferomedial migration and midline fusion, marked by the scrotal raphe	Labia majora

- The table shows that **every male structure has a female homologue**, and that the female outcome is the default whenever androgen action fails — in females, and in males with defective testosterone or DHT production, **5 α -reductase deficiency** or **androgen-receptor insufficiency**.

- The phallus develops **proximal to distal**. The long-held theory that the **glanular urethra forms by ingrowth of surface epithelium** has been challenged by evidence that it forms by **fusion of the urethral plate**.

Concept — *Male genital development is active and androgen-dependent, whereas female development is the passive default; a break anywhere in the testosterone → DHT → receptor chain therefore produces undervirilisation rather than a new structure.*

Penile Length and Tanner Classification

- Penile length rises with gestational age, from **about 6 mm at 16 weeks to about 26 mm at 38 weeks**, and prenatal measurements correlate with postnatal ones.
- A second rapid phase occurs in the **first 3 months of life (minipuberty)**: at birth the inhibitory effect of **maternal oestrogen on the fetal pituitary is lost**, producing a **gonadotrophin surge** and increased Leydig-cell testosterone. Growth then slows through childhood and accelerates again at puberty until adult size is reached.
- The normal **term neonate** has a **stretched length of 3.5 ± 0.7 cm and diameter of 1.1 ± 0.2 cm**; the adult stretched length is **13.3 ± 1.6 cm**. Stretched penile length (SPL) is measured from the **penopubic junction (pubic symphysis) to the meatus / tip of the glans**.

TABLE 55.2 STRETCHED PENILE LENGTH (CM) IN NORMAL MALE SUBJECTS

Age	Mean $\pm$ standard deviation (SD)	Mean – 2.5 SD (micropenis cut-off)
Neonate, 30 weeks' gestation	2.5 ± 0.4	1.5
Neonate, 34 weeks' gestation	3.0 ± 0.4	2.0
0–5 months	3.9 ± 0.8	1.9
6–12 months	4.3 ± 0.8	2.3
1–2 years	4.7 ± 0.8	2.6
2–3 years	5.1 ± 0.9	2.9
3–4 years	5.5 ± 0.9	3.3
4–5 years	5.7 ± 0.9	3.5
5–6 years	6.0 ± 0.9	3.8
6–7 years	6.1 ± 0.9	3.9
7–8 years	6.2 ± 1.0	3.7
8–9 years	6.3 ± 1.0	3.8
9–10 years	6.3 ± 1.0	3.8
10–11 years	6.4 ± 1.1	3.7
Adult	13.3 ± 1.6	9.3

Data from Feldman and Smith, 1975; Schonfeld and Beebe; Tuladhar et al., 1998.

- **What the table is for:** the third column is the **definition of micropenis** — an SPL more than 2.5 SD below the mean for age. The figure to remember is **1.9 cm in the term neonate and young infant**.
- **Pattern:** the mean rises only slowly through childhood (about 4 to 6.4 cm), so the cut-off **plateaus at about 3.7–3.9 cm from 5 to 11 years**; the pubertal spurt then carries the adult cut-off to 9.3 cm.
- **Traps:** the cut-off **falls slightly after 6–7 years although the mean still rises**, because the SD widens — always compare against the cut-off column, not the mean. **Preterm neonates need gestation-specific values** (1.5 cm at 30 weeks, 2.0 cm at 34 weeks).
- **Tanner stages** are the recognisable pubertal changes of pubic hair, penis and testes, used to evaluate and classify patients; they run from preadolescent (stage 1) to adult (stage 5).

TABLE 55.3 TANNER CLASSIFICATION OF SEXUAL MATURITY STAGES IN BOYS

Stage	Pubic hair	Penis	Testes / scrotum
1	None	Preadolescent	Preadolescent
2	Scanty, long	Slight enlargement; slightly pigmented	Scrotum enlarged, pink, texture altered
3	Darker, starts to curl, small amount	Longer	Larger
4	Adult type but less in quantity; coarse, curly	Larger; glans and breadth increase	Larger; scrotum darkens
5	Adult distribution, spreading to medial thighs	Adult size	Adult size

Data from Tanner, 1962.

- **What the table is for:** it provides the staging against which **testicular volume norms (Table 55.4)** and pubertal events such as **varicocele appearance (peak at Tanner 3)** and the timing of semen analysis (Tanner 5) are referenced.
- **Pattern:** stage 2 is marked by **scrotal and testicular change with only slight penile change**; the penis lengthens at stage 3 and gains girth and glans size at stage 4; the scrotum darkens at stage 4.

Concept — *Penile size is interpretable only against age-specific stretched-length norms; stretched length, not flaccid appearance, is the measurement on which the distinction between micropenis and inconspicuous penis later rests.*

PENILE ANOMALIES

Prepuce (Foreskin)

Phimosis and Paraphimosis

- At birth there is a **physiological phimosis** — partial or complete inability to retract the prepuce — caused by **natural adhesions between the glans and inner preputial skin and/or a preputial ring**.
- Two factors separate the prepuce from the glans: **(1) accumulation of smegma (epithelial debris) beneath the prepuce during the first 3–4 years**, and **(2) intermittent erections**.
- Retractability increases with age: **about 90% of uncircumcised boys have a fully retractable prepuce by 3 years, and fewer than 1% are phimotic at 17 years**. Primary phimosis therefore usually resolves during childhood.
- **Secondary phimosis** results from scarring — for example after **forceful retraction** with subsequent scarring of the prepuce to the glans — or from **balanitis xerotica obliterans (BXO)**. **Forceful retraction should be discouraged** because it causes cicatrix.
- Conditions associated with the uncircumcised penis include **paraphimosis, infection, urinary tract infection (UTI) and cancer**.
- **Paraphimosis** is entrapment of the retracted prepuce behind the glans. **Severe oedema develops within hours**, faster the tighter the preputial tip, making reduction harder; **untreated, it can progress to gangrene**. Reduction must therefore be timely, by manipulation, dorsal slit or circumcision.
- Management in escalating order: **manual compression of the glans with distal traction on the oedematous prepuce** (successful in most) → adjuncts to reduce oedema: **an iced glove for 5 minutes, granulated sugar for 1–2 hours, or multiple punctures of the oedematous skin** → dorsal slit → circumcision.

- Indications to improve retractability are **persistent primary phimosis, secondary phimosis, balanitis, posthitis (inflammation of the prepuce), BXO and UTI.**
- **Topical corticosteroid** is effective with few side-effects. **Betamethasone 0.05% twice daily for 30 days and three times daily for 21 days are equally effective**, response may be seen **within 2 weeks**, and roughly **one in six fails** — attributed to **increased collagen and reduced elastic-fibre density** in the prepuce.

Concept — *Non-retractility is normal physiology that time, smegma and erections resolve; treatment is aimed at scarring (secondary phimosis, BXO) or its consequences (infection, paraphimosis), and forceful retraction converts the physiological problem into the pathological one.*

Circumcision

- Circumcision dates back **more than 6000 years**; the oldest documented evidence is thought to be **Egyptian sixth-dynasty tomb art (2345–2181 BCE)**. Proposed origins include religious sacrifice, a rite of passage, hygiene, differentiation of cultural groups and discouragement of masturbation.
- **Elective neonatal circumcision remains controversial**, and the AAP position has evolved:

AAP POLICY ON NEONATAL CIRCUMCISION

Year	Position
1989	Potential medical benefits and advantages, as well as disadvantages and risks, to be explained to parents
1999	Emphasised the importance of local anaesthesia for the procedure
2012	Benefits (reduced UTI, reduced heterosexual acquisition of human immunodeficiency virus (HIV), reduced transmission of other sexually transmitted infections) not great enough to recommend routine circumcision of all newborns, but sufficient to justify access for families who choose it and to warrant third-party payment

- **Devices:** the **Gomco clamp, Mogen clamp and Plastibell**. Before using any device there must be **complete separation of the prepuce from the glans and full inspection of the meatus and corona to exclude anomalies, including hypospadias**. Older infants and children are usually circumcised under general anaesthesia by **freehand** techniques.
- **Local anaesthesia is recommended** for neonatal circumcision:

ANAESTHETIC OPTIONS FOR NEONATAL CIRCUMCISION

Option	Key point
EMLA cream (eutectic mixture of local anaesthetics: lidocaine and prilocaine), topical	Inferior to nerve block; prilocaine carries a low risk of methaemoglobinaemia
Dorsal penile nerve block	Superior to EMLA in randomised controlled trials
Penile ring block	Alternative infiltration technique

- **Do not circumcise neonates with penile conditions that need the intact prepuce for reconstruction or that preclude the devices:**

WHEN TO DEFER OR AVOID NEONATAL CIRCUMCISION

Category	Conditions	Reason
Anomalies needing preputial skin	Hypospadias, penile curvature, penile torsion, dorsal hood deformity, buried penis, webbed penis	Skin needed for later repair; devices unsuitable
Conditions pushing penile skin forward	Large hydrocele, inguinal hernia	Raise the risk of secondary phimosis, buried penis and trapped penis — defer
Bleeding risk	Known coagulopathy (contraindication); hyperbilirubinaemia (regarded as a contraindication by about one-third of paediatric urologists)	Haemorrhage

Category	Conditions	Reason
Setting	Neonatal intensive care unit (NICU)	Associated with more complications

- **Benefits of circumcision** are prevention of **penile cancer, UTI, sexually transmitted diseases including HIV, and phimosis**, and reduced **balanitis**. The allegation that neonatal circumcision causes **sexual dysfunction is not supported by long-term studies**, and **neonatal circumcision is more cost-effective than postneonatal** circumcision (procedure charges plus future health-care costs).
- **Penile cancer**: carcinoma of the penis develops **almost exclusively in men not circumcised at birth**, and **phimosis is a significant risk factor**. Neonatal circumcision protects much more strongly against **invasive cancer** than against carcinoma in situ.
- **UTI**: classic studies found that febrile infant UTIs occur overwhelmingly in uncircumcised boys, confirmed in larger cohorts; the excess risk lasts **at least to 5 years of age** and is attributed to **colonisation of the prepuce by uropathogens**. A **retractile prepuce halves the risk of recurrent febrile UTI** compared with a persistently non-retractile one; significant phimosis raises infant infection risk; and **topical steroid treatment of phimosis after a UTI (with normal ultrasound) significantly reduces recurrence**. It is calculated that **111 neonatal circumcisions are needed to prevent one UTI**, yet neonatal circumcision is **cheaper than treating UTIs** in uncircumcised boys. **Epididymitis** is also less common after circumcision.
- **HIV**: large randomised series show about **60% protection against HIV acquisition in men**. However, a randomised trial of **circumcising HIV-infected men showed no reduction in transmission to their female partners**. Circumcision may reduce **ulceration, bacterial vaginosis and trichomoniasis** in female partners.
- **Human papillomavirus (HPV) and herpes simplex virus type 2 (HSV-2)**: randomised trials support a protective effect, with **lower HSV-2 seroconversion and lower prevalence of high-risk (carcinogenic) HPV genotypes** after circumcision.

KEY POINTS: CIRCUMCISION

- **Primary phimosis resolves spontaneously** in the vast majority; fewer than 1% of 17-year-olds are phimotic.
- Indications to improve retractability: **persistent primary phimosis, secondary phimosis, balanitis, posthitis, BXO and UTI**.
- **Topical corticosteroid relaxes the phimotic ring in the large majority of cases**.
- The **AAP acknowledges health benefits** sufficient to justify access for families who choose it and third-party payment.
- **Local or topical anaesthesia** should be given for neonatal circumcision.
- **Defer circumcision** when another penile abnormality detected in the newborn may require surgery.
- Health benefits: reduced **penile cancer, sexually transmitted diseases including HIV, UTI, balanitis and phimosis**.

Concept — *The benefits of circumcision all follow from removing a moist, colonisable and potentially phimotic preputial sac; the AAP's position is a cost-benefit judgement that supports access without mandating the procedure, and the prepuce is preserved whenever it may be needed for reconstruction.*

Glanular adhesions and skin bridges.

- Both are attachments of the glans to the shaft, **common after circumcision** and usually noticed by a carer. They occur regardless of how complete the circumcision was, and result from **physiological retraction of the penis into the suprapubic fat pad, diaper irritation, or incomplete lysis of adhesions at circumcision**.

- A **glanular adhesion** joins the inner prepuce or circumcision line to the glans. Its **incidence falls steadily with age** through epithelial separation — common in infancy, rare after 9 years. Persistent adhesions can be **lysed in the office** after a topical analgesic (EMLA or a vapocoolant spray). **Low-dose corticosteroids are not supported.**
- A **skin bridge** is an **epithelialised adhesion** that can cause **chordee and torsion**. It is excised in the office with analgesia, or in theatre when broad or thick and requiring haemostasis or suturing of the glanular and shaft defects.
- The **commonest acquired penile lesion is entrapped smegma** beneath a non-retractile prepuce or glanular adhesions — a **yellowish mass** that is released as the adhesions resolve spontaneously or are lysed.

Meatal stenosis.

- In children meatal stenosis occurs **almost only after circumcision in infancy**. It may be **congenital, mainly in neonates with hypospadias**, or acquired.
- Normal meatal calibre by age (calibrate with a **bougie à boule or urethral sound**):

NORMAL MEATAL CALIBRE BY AGE

Age	Calibre (French, Fr)
Neonates under 4 months	6–10 Fr
Infants 1–12 months	10–12 Fr
Children 1–10 years	12–14 Fr
Adolescents over 10 years	14–18 Fr

- Proposed causes of secondary stenosis: **severe meatal inflammation and cicatrix after lysis of normal adhesions and circumcision, frenular devascularisation, and chronic meatitis from diaper irritation of the exposed meatus**. **BXO** is another cause. Only a small proportion of circumcised boys become symptomatic.
- The three symptoms: **(1) upward deviation of the stream from a meatal baffle or ventral web** at the inferior meatus; **(2) a narrow, high-velocity stream** with a sense of incomplete emptying, straining or frequency; **(3) pain at the start of micturition**.
- Urethrography (voiding or retrograde) may show proximal urethral dilatation but is **not needed for diagnosis**; reserve it for **UTI, incontinence or suspected more proximal urethral abnormality**.
- **Meatotomy or meatoplasty** — in the office with topical EMLA, or under general anaesthesia — uses a **ventral incision long enough to restore normal calibre**. **Suturing the urethral mucosa to the glans with fine, rapidly absorbable sutures reduces recurrence**; **clamp tailoring without sutures is equally effective**.

Penile trauma.

- The **most serious circumcision complications** are **urethral injury, excision of the glans and/or shaft, and penile necrosis**.
- Urethral injury requires **urethroplasty** matched to the severity of excision. An **excised glans can be sutured back**, often **without microscopic repair**, with good results **if done within 8 hours**.
- **Penile necrosis** follows **thermal injury** — for example **cautery touching the metal ring or clamp**, or inappropriate use of lasers. The options are all imperfect: **penile reconstruction, or female gender reassignment with bilateral orchidectomy**; the latter is problematic because many such children **grow up with a male identity, attributed to prenatal and neonatal androgen imprinting**.

Circumcision complications.

- The overall complication rate is **0.2–5%**. It is low when circumcision is performed in the first year of life but **rises 10- to 20-fold after infancy**. Complications may present immediately or months to years later.
- **Bleeding is the commonest complication (about 0.1%)**, more frequent in older children. It usually arises from the **frenulum**, less often from a shaft vessel or a skin edge between sutures, and is managed with **compression, ophthalmic cautery, a silver nitrate stick or a suture**; occasionally exploration is needed to evacuate a haematoma.
- **Wound infection is rare; antibiotic ointment (e.g., bacitracin)** usually prevents it. Rare cases of **Fournier gangrene** are reported.
- **Penile degloving** — from excessive skin removal or failure of the skin edges to adhere to the mucosal collar — **re-epithelialises without intervention**, using antibiotic ointment and warm baths to prevent eschar. **Immediate suturing and skin grafting are not recommended.**

Penile skin complications.

- **Insufficient or asymmetrical excision** causes health, cosmetic and social problems as the child grows. **Revision requires general anaesthesia**, unlike neonatal circumcision.
- **Excessive excision** can cause **chordee, torsion and lateral deviation**, repaired if necessary with **penile skin flaps, Z-plasty or split-thickness grafts**, and a **trapped penis** from a cicatricial scar.
- A trapped penis is managed with **topical betamethasone 0.05% plus manual retraction**, which softens the cicatrix in most children, exposing the glans or leaving a mild ring amenable to a **vertical relaxing incision**; otherwise formal repair with penile skin flaps or Z-plasty.

Balanitis xerotica obliterans.

- **BXO — lichen sclerosus et atrophicus** — is a **chronic infiltrative and cicatrising skin condition that causes pathological phimosis**. It can affect the **glans, meatus and urethra**.
- The usual presentation is **inability to retract the prepuce at puberty; BXO is rare before 5 years**. Other symptoms are local infection, irritation, discomfort after micturition, bleeding, and occasionally acute retention or incontinence. **Older patients and those with meatal involvement have a more severe course.**
- The cause is unknown — no viral, bacterial or familial cause has been identified — and the **association with penile cancer in children is not established.**
- **Topical corticosteroids help only mild preputial disease** with minimal scarring. **Circumcision is the preferred treatment, with meatotomy or meatoplasty when the meatus is involved.** Children with meatal disease must be followed because **about one-fifth need further meatal surgery**. **Buccal mucosal urethroplasty** is promising in severe disease.
- A **prepuce-sparing approach — triradiate preputial incisions with intralesional triamcinolone** — left most boys with a fully retractile prepuce free of visible disease, with **significantly less meatal stenosis than primary circumcision**; a minority developed recurrence requiring circumcision or repeat preputioplasty.

KEY POINTS: CIRCUMCISION COMPLICATIONS

- The complication rate is **0.2–5%**; **bleeding is the most common.**
- **Circumcision revision requires general anaesthesia.**
- **Meatal stenosis occurs almost exclusively in the circumcised penis.**
- **Topical corticosteroids do not resolve penile adhesions.**

- **BXO is a chronic infiltrative and cicatrising skin condition.**

Concept — *The complication rate tracks age at circumcision, and most late sequelae are either problems of skin quantity (too much removed → trapped penis, chordee, torsion; too little → redundancy and adhesions) or problems of the newly exposed meatus (stenosis).*

Abnormal Penile Number

Aphallia

- **Penile agenesis results from failure of development of the genital tubercle.** It is very rare — **about 1 in 10–30 million births.**
- The **karyotype is almost always 46,XY.** The usual appearance is a **well-developed scrotum with descended testes and no penile shaft**; the **anus is usually displaced anteriorly**, and the urethra opens at the anal verge beside a small skin tag, or into the rectum.
- **About one in three cases ends in stillbirth or neonatal death.**
- Associated malformations are common: **cryptorchidism, vesicoureteral reflux (VUR), horseshoe kidney, renal agenesis, imperforate anus, and musculoskeletal and cardiopulmonary anomalies.** The genitourinary–gastrointestinal connection varies, and its level predicts outcome:

APHALLIA BY LEVEL OF THE URETHRAL OPENING (SKOOG AND BELMAN CLASSIFICATION)

Urethral opening	Relative frequency	Survival	Associated anomalies
Postsphincteric — meatus on an appendage at the anal verge	Commonest	Best	Fewest
Presphincteric — prostaticorectal fistula	Intermediate	Intermediate neonatal mortality	Intermediate
Urethral atresia with vesicorectal fistula for drainage	Least common	Universally fatal in the series	Most

- **Rule:** the **more proximal the urethral opening, the higher the neonatal mortality and the more associated anomalies.**
- Aphallia should be **considered in apparent females with severe urinary tract dysplasia and genital ambiguity**, with imaging of the urinary tract (ultrasound and cystography).
- **Evaluate immediately and in a multidisciplinary team:** **karyotype**, studies for associated malformations of the urinary tract and other systems, and **magnetic resonance imaging (MRI)** to define the severity of the defect.
- **Advances in neophallus reconstruction now offer an alternative to remaining aphallic or to gender reassignment.** Raised as a male the patient would potentially be fertile; phalloplasty and urethroplasty allow sexual and voiding function, although reproductive function is unknown.
- **Gender reassignment should be recommended only after full multidisciplinary evaluation and parental counselling**, because some patients reconstructed as female **develop a male gender identity**, presumably from **in utero or postnatal sex-steroid imprinting**. Reassignment involves **neonatal orchidectomy and feminising genitoplasty, with a neovagina later**; combined urinary tract reconstruction with an **intestinal neovagina through a posterior sagittal and abdominal approach** has been described.

Diphallia

- **Penile duplication is rare** — **about 1 in 5 million live births** — and ranges from a **small accessory penis to complete duplication.**

- Each phallus may have **one corporal body and a urethra**, or — as a variant of twinning — **two corpora cavernosa and a urethra**. The penises are usually **unequal in size and adjacent**.
- Associated anomalies are common: **hypospadias, bifid scrotum, bladder duplication, renal agenesis or ectopia, and diastasis of the pubic symphysis**, plus anal and cardiac anomalies.
- **Evaluate the entire urinary tract with renal ultrasound and a voiding cystourethrogram (VCUG)**; ultrasound and MRI assess penile development. The cause is unknown, and **treatment is individualised** around the associated anomalies and a satisfactory functional and cosmetic result.

APHALLIA VERSUS DIPHALLIA

Feature	Aphallia	Diphallia
Incidence	1 in 10–30 million births	1 in 5 million live births
Defect	Genital tubercle fails to develop	Duplication; spectrum from accessory penis to twinning
Karyotype / gonads	46,XY ; scrotum and descended testes normal	Not specified
Typical associations	Anorectal (anterior anus, rectal urethral opening), VUR, renal, skeletal, cardiopulmonary	Bladder duplication, bifid scrotum, renal, pubic diastasis, anal, cardiac
Key evaluation	Karyotype, anomaly screen, MRI	Renal ultrasound + VCUG, ultrasound/MRI of penis
Decision	Neophallus reconstruction versus reassignment (multidisciplinary)	Individualised reconstruction

Concept — *In both anomalies the associated malformations, not the phallus, determine survival and dictate the work-up; the more proximal or extensive the defect, the heavier the associated burden.*

Inconspicuous Penis

- An **inconspicuous penis appears small but has a normal stretched length** (pubic symphysis to glans tip, Table 55.2) **and a normal shaft diameter**. It is congenital or acquired and causes great parental concern.
- The group includes **buried penis, trapped penis and webbed penis**, all of which **must be distinguished from micropenis**, in which the penis is truly small. Evaluation is needed both for treatment and to counsel the family on whether the penis is normal.

INCONSPICUOUS PENIS VERSUS MICROPENIS

Feature	Inconspicuous penis (buried, trapped, webbed)	Micropenis
Stretched length	Normal	≥ 2.5 SD below mean
Nature of problem	Skin, fat, fixation or scar concealing a normal shaft	Small, normally formed organ
Cause	Anatomical (dartos, obesity, cicatrix, web)	Hormonal (gonadotrophin or testicular deficiency)
Management	Treat the anatomical cause; surgery if indicated	Endocrine evaluation; androgen therapy

Buried Penis

- A **buried (hidden, concealed) penis is a normally developed penis hidden by the suprapubic fat pad**. It is classified by the cause of concealment:

BURIED PENIS — THE THREE CATEGORIES

Category	Mechanism	Treatment
(1) Poor penopubic and/or penoscrotal skin fixation (congenital)	Inelastic dartos: normally the dartos lets the skin slide on the deep layers; here the skin is not anchored to the deep fascia , restricting extension	Surgery (see below)
(2) Obesity (acquired; older children and adolescents)	Abundant abdominal-wall fat hides the shaft	Treat the obesity — refer to primary care for weight loss and exercise, then reassess the need for surgery
(3) Trapped penis (acquired; after penile surgery, typically circumcision)	Cicatrix over the glans embeds the penis in the suprapubic fat	Office dilation of the cicatrix with a fine haemostat under analgesia, or betamethasone 0.05% + manual retraction (resolves most or permits simple incision, largely avoiding formal repair) → if ineffective, repair under general anaesthesia at ≥ 6 months , as for congenital buried penis

- Trapped penis may follow **neonatal circumcision in an infant with significant scrotal swelling from a hernia or hydrocele, routine circumcision of a webbed penis**, or circumcision when the shaft **naturally retracts into the scrotum** — the skin at the base then forms a cicatrix over the retracted phallus.
- Distinguish a **transient buried penis** from a large suprapubic fat pad in early childhood, which **resolves with age, walking and shrinkage of the fat pad**.
- **On examination the buried penis must be distinguished from micropenis by its normal stretched length.** Determine whether the **glans can be exposed by retracting the overlying skin** (retracting the skin lateral to the shaft reveals it); if so, whether to correct is the surgeon's judgement.
- If the penis is trapped — by **physiological phimosis in the uncircumcised penis or cicatrix in the circumcised penis** — there is an increased risk of **difficult voiding, poor hygiene, balanitis, UTI (from urine pooling) and psychosocial problems**.
- **Surgical indications and timing are controversial.** Many techniques exist. In moderately severe cases the sequence is: **excise dysgenic bands (mainly on the proximal dorsal shaft) → unfurl the prepuce for ventral skin cover → fix the dorsal penile subcutaneous tissue to the pubic fascia → fix the scrotal subcutaneous tissue to the ventral base of the shaft**, taking care to **avoid the neurovascular bundles**.
- Skin cover may instead use a **penoscrotal Z-plasty, lateral shaft Z-plasty, island pedicle of ventral preputial skin, or a skin graft**. Rarely, in the most severe cases, the **suspensory ligament is divided, the suprapubic fat excised, and the spermatic cords protected**.
- **Liposuction** has benefit in severe cases but should be **reserved for adolescents**, because **prepubertal children may lose the fat pad simply with growth**.

Webbed Penis

- A **webbed penis (penoscrotal fusion)** is scrotal skin extending onto the ventrum of the penis. The **congenital** form is an abnormal penoscrotal attachment with a **normal penis, urethra and remaining scrotum**; the **acquired** form follows **excessive removal of ventral skin** at circumcision or other penile surgery.
- It is **usually asymptomatic but cosmetically unacceptable**. Repairs: **fixation of scrotal subcutaneous tissue to the ventral base of the shaft** (as for buried penis); **transverse incision of the web, separation of penis from scrotum, and vertical closure**; or a **circumferential incision 1.5 cm proximal to the coronal**

sulcus, transfer of a Byars' preputial flap to the ventrum, and excision of redundant foreskin. Rarely the **distal urethra is hypoplastic** and needs reconstruction.

Concept — *Inconspicuous penis is a problem of skin, fat or fixation on a normal-sized organ, whereas micropenis is a problem of organ size with an endocrine cause; stretched length is the single discriminator, and treatment of each inconspicuous form is directed at its specific cause.*

Micropenis

- **Micropenis is a normally formed penis at least 2.5 SD below the mean stretched length for age; in the term neonate this means under 1.9 cm.** The length-to-circumference ratio is usually normal, though the corpora are occasionally hypoplastic. The **testes are usually small and often cryptorchid**, and the scrotum is usually fused and often diminutive.
- **Stretched length correlates more closely with erect length than relaxed length does.** Measure from the pubic symphysis to the tip of the glans with **complete compression of the suprapubic fat pad**, especially in an obese child, and compare with Table 55.2. **Buried and webbed penis must be excluded — both have a normal shaft.**
- **Embryology explains the phenotype:**

HORMONAL TIMING OF PENILE DEVELOPMENT

Period	Driver	Defect produces
To week 12 — differentiation complete	Testicular testosterone stimulated by placental human chorionic gonadotrophin (hCG)	Defective testosterone production or action → small penis with hypospadias
Second and third trimesters — growth	Fetal androgen under fetal luteinising hormone (LH)	Gonadotrophin deficiency (after 14 weeks) → true micropenis, normally formed

- **The commonest causes are hypogonadotrophic hypogonadism (the single most common), hypergonadotrophic hypogonadism (primary testicular failure) and idiopathic micropenis.** Micropenis is often associated with major chromosomal defects: **Klinefelter syndrome (47,XXY)** and other X polysomies, deletions, translocations, and **trisomies of chromosomes 8, 13 and 18.**

CAUSES OF MICROPENIS BY LEVEL OF THE AXIS

Level	Mechanism	Examples	Laboratory signature
Hypogonadotrophic hypogonadism (commonest)	Hypothalamus fails to produce adequate gonadotrophin-releasing hormone (GnRH)	Prader-Willi, Kallmann (genital-olfactory dysplasia), Laurence-Moon-Biedl, CHARGE association; growth hormone deficiency or neonatal hypoglycaemia from congenital hypopituitarism; congenital pituitary aplasia; midline brain defects (agenesis of corpus callosum, occipital encephalocele)	Low LH and FSH; in Kallmann syndrome with cryptorchidism testosterone may not rise after hCG
Hypergonadotrophic hypogonadism (primary testicular failure)	Testis cannot respond	Rudimentary testes syndrome, gonadal dysgenesis, Robinow syndrome	Testosterone fails to rise after hCG; elevated basal LH and follicle-stimulating hormone (FSH)
Idiopathic	Improper timing or delayed onset of fetal gonadotrophin stimulation	—	Axis normal

Level	Mechanism	Examples	Laboratory signature
Androgen resistance (rare)	Partial androgen insensitivity syndrome	—	Ambiguity is more usual than micropenis

- **Evaluation** in sequence: **thorough history, examination and a karyotype at birth**; accurate SPL, palpation of the corpora, and assessment for cryptorchidism → **paediatric endocrinologist** to determine central versus testicular cause, detect other abnormalities and judge growth potential → **serum testosterone before and after hCG stimulation** ($\pm$ GnRH stimulation test) → **anterior pituitary screen: serial glucose, sodium and potassium, cortisol, thyroid function** → **MRI of the head** for the hypothalamus, anterior pituitary and midline structures.
- **Before extensive axis evaluation, give androgen to test end-organ response**: usually **intramuscular testosterone enanthate for 3 months**. Prolonged treatment may advance skeletal maturation, but **short courses do not affect height**; transdermal testosterone has also been used.
- Whether neonatal androgen that brings the penis into the normal range affects pubertal growth is unclear; treatment **at birth and again at puberty** produced normal adult stretched length in a small series, so **testosterone enanthate at both times is recommended** until longer-term data exist.
- **Outcome**: although final size may remain below normal, individuals born with micropenis have a **male gender identity, stand to void, and most have satisfactory sexual function** and are sexually active; some remain dissatisfied with genital appearance and function, and parents worry about future size and function.

KEY POINTS: MICROPENIS

- **Micropenis is a normally formed penis at least 2.5 SD below the mean stretched length for age (under 1.9 cm in neonates).**
- **Buried penis is not micropenis** — its stretched length is normal.
- The commonest causes are **hypogonadotrophic and hypergonadotrophic hypogonadism**.
- **Evaluation: karyotype; serum LH and FSH; testosterone before and after hCG; anterior pituitary function; MRI of the brain** (hypothalamus, anterior pituitary, midbrain).
- In adulthood most have a **male gender identity and satisfactory sexual function**.

Concept — *Micropenis is an endocrine diagnosis made with a ruler: a normally formed but small penis places the defect after differentiation is complete, in gonadotrophin-driven growth, so the work-up targets the hypothalamic–pituitary–testicular axis and treatment first tests the end-organ response to androgen.*

Abnormal Penile Orientation

Penile Curvature

- Curvature may be **vertical (ventral or dorsal) or horizontal (lateral)**, **congenital or acquired** after circumcision, hypospadias repair or trauma, and affects cosmesis, body image and later sexual function.
- **Indications for repair: known associated sexual dysfunction, concurrent penile surgery, and the surgeon's judgement that worsening curvature with growth will cause sexual dysfunction.**

PENILE CURVATURE BY DIRECTION

Feature	Ventral (chordee)	Dorsal	Lateral
Frequency	Commonest	Less common	Less common

Feature	Ventral (chordee)	Dorsal	Lateral
Associations	Hypospadias ; isolated chordee ± dorsal hooded prepuce, deficient ventral skin ; prominent frenulum → distal chordee with ventral glans deflection	Isolated ± asymmetric skin; epispadias with ventral hood ; some have hypospadias	Usually congenital — overgrowth or hypoplasia of one corpus ; secondary to asymmetric skin excision or scarring
Recognition	At birth / with hypospadias	At birth	Often not recognised until later childhood — straight when flaccid, curved only when erect
First-line repair	Deglove → excise fibrous tissue superficial to Buck's fascia → Byars' flaps ; catheter protects urethra	Deglove → excise superficial fibrous tissue → skin flaps; protect the dorsal neurovascular bundles	Deglove → plication and/or ellipse excision of the corpus contralateral to maximum curvature
Escalation	Dorsal plication, Nesbit dorsal excision or corporal rotation → short urethra: urethral reconstruction ; frenular chordee: frenulotomy	Plication and/or ellipse excision of the ventral corpora	Secondary lateral curvature managed as vertical curvature

- **Intraoperative artificial erection** — saline injected into a corporal body with or without a tourniquet — **identifies the apex and confirms complete correction** in every type.

Concept — *Correction always proceeds from superficial to deep: release the skin and superficial fibrous tethers first (the commonest cause), then, only if curvature persists, shorten the convex side by plication or ellipse excision — dorsal plication for ventral curvature, ventral for dorsal, and the contralateral corpus for lateral.*

Penile Torsion

- **Penile torsion is a rotational deformity of the shaft, usually counter-clockwise (to the left).** Penile size is usually normal, and the condition is often **unrecognised until circumcision or preputial retraction**. It may accompany **hypospadias, chordee or dorsal hood deformity without urethral abnormality**.
- The **median raphe usually spirals obliquely** in the direction of the torsion and inserts atypically rather than at the base of the glans in line with the meatus. The cause is probably an **anomalous arrangement of penile shaft skin**.
- **Most torsions are less than 60°.** If repair is being considered, **neonatal circumcision is discouraged and is done at the time of detorsion**. Even when the glans points more than 90° from the midline, **the corpora and corpus spongiosum are usually normally oriented at the base**.
- **Mild torsion:** deglove → **rotate the glans opposite to the defect (clockwise for a counter-clockwise torsion)** → suture the glans to the penile skin. A serpentine raphe that cannot be restored is acceptable **provided the meatus is normal**.
- **Torsion of 90° or more:** skin rearrangement alone is insufficient → **mobilise the base and incise dysgenic fibrous bands** → if still rotated, place **non-absorbable anchoring sutures at the base of the corpora cavernosa**, possibly rotating a **dorsal dartos flap** or mobilising the **urethral plate and urethra**.

Concept — *Torsion is a skin-level deformity on normally oriented corpora in all but severe cases, so degloving and redraping suffice below 90°, and only beyond that must the base be mobilised and anchored.*

Penile Masses

- **Cysts, congenital or acquired, are the commonest penile masses in children.** The history of onset, previous surgery and change in size or appearance are central to evaluation.

PENILE MASSES IN CHILDREN

Lesion	Features	Origin	Management
Parameatal urethral cyst	Rare; small blister beside the meatus ; wall transitional and squamous or columnar	Occlusion of paraurethral ducts, or faulty preputial separation along the coronal sulcus	Complete excision and marsupialisation of the urethral mucosa , avoiding meatal stenosis
Median raphe cyst	Congenital epidermal cysts along the raphe — glans, shaft, scrotum or perineum; rarely extend into the pelvis	Epithelial rests buried during urethral infolding, or a monodermal teratoma	Excise under general anaesthesia unless small and asymptomatic
Epidermal inclusion cyst	After circumcision or hypospadias repair	Islands of epithelium trapped in subcutaneous tissue	Excise — enlarge, distort skin, cause discomfort during sexual activity
Congenital penile naevus	Pigmented lesion of glans or shaft; dermal, junctional or compound by melanocyte location	Congenital	Superficial and benign; excise for definitive diagnosis
Juvenile xanthogranuloma	Uncommon, benign, self-limiting ; infancy/early childhood; solitary or multiple yellow, orange, gold, brown or red nodules of rapid onset, 2–20 mm , well demarcated, firm, rubbery; penis or scrotum; up to one-fifth present at birth	—	Observe for 1 year to avoid unnecessary ablative surgery

NAEVUS CLASSIFICATION BY MELANOCYTE LOCATION

Type	Melanocytes
Dermal	Dermis only
Junctional	Dermal-epidermal junction only
Compound	Dermis and dermal-epidermal junction

Concept — *Management follows natural history: lesions that grow or need histology (inclusion cysts, naevi, symptomatic raphe cysts) are excised, whereas the self-limiting juvenile xanthogranuloma is watched.*

Accessory Urethral Openings**Congenital Urethral Fistula**

- The urethra and meatus are normal, with a **urethrocutaneous fistula, typically coronal or subcoronal**. It is **usually isolated** but may accompany **imperforate anus or ventral chordee**; distal hypospadias and chordee occur in a minority.
- The cause is unknown — possibly a **focal urethral-plate defect preventing fusion of the urethral folds**. Some cases are diagnosed after circumcision, raising the possibility of an **iatrogenic injury**.
- Repair: **circumscribe the fistula and close in multiple layers**, as for a post-hypospadias fistula; or, **if the glans bridge is thin, open the ventral glans through the distal urethra and repair by Thiersch–Duplay tubularisation** with urethral-plate incision as needed.

Urethral Duplication

- **Rare** (about 200 reported cases). **Duplication is most commonly sagittal, one urethra ventral and one dorsal (about 94%)**; collateral duplication in the same horizontal plane is **extremely rare**.
- The **ventral urethra is usually the functional one** — it carries the stream and the landmarks (**external sphincter, verumontanum**); the **dorsal urethra is the accessory**, with or without a stream.

- **Effman classification (1976)**, the most widely used: **Type I partial (incomplete)** and **Type II complete** duplication; **Type III duplication as part of bladder duplication**. The figure subdivides these into IA, IB, IIA1, IIA2 (including the "Y-type") and IIB.
- The embryology is not established. Associated genitourinary, gastrointestinal and musculoskeletal anomalies are frequent — **VUR is the commonest**; others include renal agenesis, bilateral cryptorchidism, sacral agenesis, imperforate anus, radial hypoplasia and tracheo-oesophageal fistula — with an association with **midline defects** (duplicated bladder or colon, anorectal agenesis, bifid glans, thoracic hemivertebrae, partial sacral agenesis).
- **Commonest presentation: double meatus and double stream**; also **incontinence and recurrent UTI**; less often chordee, or **obstruction from a mucosal flap at the bifurcation acting as a valve**. **The more proximal the origin of the accessory urethra, the higher the incidence and degree of incontinence.**
- **Evaluate with VCUG, retrograde urethrogram and cystourethroscopy**, plus assessment of associated anomalies.
- **Treat the symptomatic child** (some accept cosmesis alone as an indication). **Never use the accessory urethra as the primary urethra — it is hypoplastic and flow may be inadequate**. Options depend on anatomy: **complete excision of the accessory tract, fulguration or sclerosant injection, septotomy if the septum is thin, or urethro-urethrostomy into the functional urethra.**

Concept — *The ventral channel is the true urethra: every rule — which channel to keep, which to ablate, and why incontinence rises with a more proximal origin — follows from the dorsal accessory tract being hypoplastic and bypassing the sphincter.*

Genital Lymphoedema

- **Lymphoedema of the genitalia is a disfiguring disorder of impaired lymphatic drainage causing progressive penile or scrotal swelling**, congenital or acquired.
- **Congenital lymphoedema is sporadic in about 85% and inherited in about 15%**, and may appear at various ages:

FORMS OF CONGENITAL LYMPHOEDEMA

Form	Inheritance	Onset
Milroy disease	Autosomal dominant	Congenital
Meige disease	Probably autosomal dominant, variable penetrance	First or second decade
Lymphoedema praecox	Sporadic	At puberty — about 80% of congenital lymphoedema presents at puberty

- Lymphoedema may involve the penis, scrotum or both, sometimes the lower limbs, and is associated with **Turner, Noonan and Klinefelter syndromes and intestinal lymphangiectasia**.
- **Crohn disease** has extraintestinal ("metastatic") manifestations in up to a third of patients. **Anogenital granulomatosis** is a rare, delayed manifestation: lesions are most commonly **oedema (mainly penoscrotal)**, ulcers and fissures. **Children present earlier and more often have luminal Crohn disease** than adults. Treatment — **antibiotics, corticosteroids, thiopurines and anti-tumour necrosis factor (TNF) agents** — produces a response in most but remission in a minority, and **males remit less often**. **Intravenous infliximab**, an anti-TNF monoclonal antibody used for moderate-to-severe Crohn disease and other inflammatory disorders, succeeded in an adolescent.
- **Surgery is considered if lymphoedema is significant or progressive; the goal is to remove all involved tissue**. On the shaft: **deglove and excise all tissue between Buck's fascia and the skin, plus redundant skin**; an uncircumcised prepuce may be circumcised or **unfurled to cover the shaft**. In the scrotum: **excise**

the scrotal skin except the posterior skin, preserving cords and testes. Cover the penis with **local skin flaps** and the scrotal contents with **uninvolved posterior flaps**, using **split-thickness grafts** if healthy skin is inadequate.

- **Counsel parents that lymphoedema may recur in adjacent areas.**

Concept — *Surgery for genital lymphoedema is excisional rather than drainage: all abnormal tissue superficial to Buck's fascia (and all but the posterior scrotal skin) must go, and cover is borrowed from uninvolved skin.*

Priapism

- **Priapism is a prolonged, commonly painful erection lasting more than 4 hours without physical or psychological stimulation.** It is uncommon in children. **Tumescence is usually confined to the corpora cavernosa**; sometimes the corpus spongiosum and glans are involved.
- Evaluation and management follow the adult approach, adapted for children, as in the guidelines of the **American Urological Association (AUA)** and **European Association of Urology (EAU)**. After history and examination, **blood gas analysis of corporal aspirate categorises priapism**:

TYPES OF PAEDIATRIC PRIAPISM

Feature	Ischaemic (veno-occlusive, low-flow)	Non-ischaemic (arterial, high-flow)	Stuttering (intermittent)	Neonatal
Mechanism	Little or no cavernous flow	Unregulated cavernous arterial inflow — cavernous artery–corpus fistula	Recurrent ischaemic episodes, shorter, with intervening detumescence	Idiopathic, birth trauma, polycythaemia
Blood gas	Hypoxic, hypercapnic, acidotic ; dark blood	Bright red ; resembles arterial blood	As ischaemic during episodes	—
Examination	Corpora rigid and tender ; pain = ischaemia	Neither fully rigid nor painful	Painful, short erections	—
Commonest paediatric cause	Homozygous sickle cell disease ; also leukaemia, other haemoglobinopathies, local malignancy	Perineal trauma (straddle injury) ; also Fabry disease, sickle cell anaemia	Sickle cell disease — commoner than prolonged priapism	—
Imaging	—	Colour Doppler ultrasonography (CDUS) often shows the fistula	—	—
First line	Hydration, oxygenation, alkalinisation, analgesia (± transfusion) → soon after, aspiration of dark blood and saline irrigation ± intracavernous α-adrenergic agonist (phenylephrine or epinephrine)	Observation — spontaneous resolution common; aspiration/irrigation is diagnostic	Oral pseudoephedrine (α-agonist) at bedtime	No intervention — resolves in 2–6 days
Next line	Surgical shunt for recalcitrant cases	Superselective embolisation of cavernous/penile arteries	Oral β-agonist ; long-term oral phosphodiesterase type 5 (PDE5) inhibitor (preliminary)	—

- **Sickle cell mechanism:** priapism occurs in **2–29% of males with sickle cell disease**. During a normal erection, **sickled red cells stagnate in the corporal sinusoids**; the resulting **fall in pH and oxygen tension**

promotes further stasis and sickling — a self-perpetuating cycle. Episodes typically occur **during sleep**, when mild **hypoventilatory acidosis** lowers corporal oxygen tension and pH, or with oxyhaemoglobin desaturation.

- Conservative and corporal treatment together succeed without later surgery in most cases. The rationale for **long-term PDE5 inhibitors** is the hypothesis that **PDE5 dysregulation contributes to priapism**.
- **Shunts bypass the veno-occlusive mechanism to drain the corpora:**

SHUNT PROCEDURES FOR RECALCITRANT ISCHAEMIC PRIAPISM

Shunt type	Named procedures
Distal cavernoglanular	Winter, Ebbehøj, Al-Ghorab
Proximal cavernospongiosal	Quackels, Sacher
Saphenous vein to corpus cavernosum	Grayhack

- **Paediatric evidence for shunt efficacy is scant;** parents must be counselled about **cavernosal fibrosis and erectile dysfunction**.

KEY POINTS: PRIAPISM

- Priapism is **ischaemic (veno-occlusive, low-flow)**, **non-ischaemic (arterial, high-flow)** or **stuttering (intermittent)**.
- **The commonest cause of priapism in children is sickle cell disease.**
- Initial treatment of sickle low-flow priapism is **conservative** — **hydration, oxygenation, alkalinisation and transfusion**; **evacuation and irrigation of the corpora with intracavernous α -adrenergic agents** may be done concurrently or after conservative failure.
- **High-flow priapism usually follows perineal trauma (straddle injury); embolise if spontaneous resolution fails.**
- **Transient erections under 2 hours or stuttering priapism are treated with bedtime pseudoephedrine.**

Concept — *The corporal blood gas divides priapism by mechanism: ischaemic priapism is a compartment emergency requiring evacuation, whereas arterial priapism is a well-oxygenated fistula that can safely be observed — so the aspirate, not the appearance, decides the urgency.*

Penoscrotal Transposition (Scrotal Engulfment)

- Transposition is **partial or complete**; lesser forms are the **doughnut, prepenile and shawl scrotum**. The embryological cause is **incomplete or failed inferomedial migration of the labioscrotal swellings**. Two- and three-dimensional ultrasound has detected it **in utero**.
- **It frequently accompanies perineal, scrotal or penoscrotal hypospadias with chordee**, and is also associated with **caudal regression, sex chromosome abnormalities and Aarskog-Scott syndrome**.
- **Up to 75% of patients with complete transposition and an otherwise normal scrotum have a significant urinary tract abnormality** (renal agenesis, dysplasia), and non-genitourinary anomalies occur — so **evaluate with renal ultrasound and VCUG**.
- **Repair is usually done with hypospadias or other penile surgery**. The aims are to **free the penis from the scrotum, develop skin flaps, relocate the scrotum dependently and move the penis superiorly**.
- **Rotation-flap scrotoplasty**: circumscribe the superior aspect of each hemiscrotum and extend the incisions laterally to include **at least half of the scrotum** → join the medial ends on the ventral penis and carry the incision down the midline raphe → infiltrate **lidocaine with epinephrine** to develop planes and reduce bleeding → deepen **without injuring the tunica vaginalis and spermatic cord** → **rotate the scrotal**

wings medially beneath the penis and suture them in the midline, everted. Ventral shaft skin is usually adequate; in severe cases **dorsal interposition flaps allow caudal advancement of abdominal skin.**

- **Buttonhole (penile relocation) alternative:** identify the correct penile position → excise a **plug of epidermis, dermis and suprapubic fat** → deglove → bring the shaft through the buttonhole → split the retained shaft skin ventrally and mobilise it superiorly → create a **window in the dartos pedicle** and pass the shaft through it → resurface with shaft skin.

Concept — *Transposition is a failure of scrotal migration rather than a penile anomaly, so repair moves the scrotal skin (or re-sites the penis through the skin) rather than altering the phallus, and its strong association with upper-tract anomalies makes imaging part of every work-up.*

SCROTAL ANOMALIES

SCROTAL ANOMALIES COMPARED

Anomaly	Defining feature	Key association	Management
Bifid scrotum	Labioscrotal folds completely separated, no median raphe	Proximal hypospadias	Repair with hypospadias repair ; technique as for penoscrotal transposition, or single or multiple Z-plasties
Ectopic scrotum	One hemiscrotum anomalously sited: suprainguinal (commonest) , infrainguinal or perineal	Cryptorchidism, inguinal hernia, bladder exstrophy, popliteal pterygium syndrome ; ipsilateral upper-tract anomalies	Upper-tract ultrasound ; scrotoplasty and orchidopexy at 6–12 months (earlier if other surgery needed)
Scrotal hypoplasia	Underdevelopment of one or both sides	Undescended testis; genital ambiguity	—
Scrotal agenesis	Absent scrotum (rare) or hemiscrotum (two cases)	All reported cases XY, bilateral undescended testes, median raphe present , other anomalies	—
Scrotoschisis	Congenital rupture of the scrotum , testis(es) extruded	Meconium peritonitis (also meconium hydrocele)	Investigate for meconium peritonitis; exploration, orchidopexy, primary closure

- **Ectopic scrotum:** suprainguinal ectopia is frequently accompanied by **ipsilateral renal agenesis, renal dysplasia or ectopic ureter**. Most affected children have an **associated perineal lipoma**; those with a lipoma usually have no other anomaly, whereas those without one have genital or renal malformations. **All should have upper urinary tract ultrasound.** Because the gubernaculum and scrotum develop together in time and place, ectopia probably reflects a **gubernacular defect preventing migration of the labioscrotal swelling**.
- **Scrotal hypoplasia** may result from a **lack of gubernacular swelling**.
- **Scrotal agenesis:** the associated anomalies include cognitive impairment, bilateral nystagmus, cleft palate and syndactyly. In **hemiscrotal agenesis** one case had multiple anomalies and one was isolated; the penis and contralateral hemiscrotum were normal, and the testis on the affected side lay in what would have been its normal position.

Concept — *Most scrotal anomalies (transposition, bifid, ectopic and hypoplastic scrotum) are failures of gubernacular-guided labioscrotal migration and fusion, so they travel with hypospadias, cryptorchidism and upper-tract anomalies — the imaging, not the scrotoplasty, is the urgent part of management.*

VASCULAR LESIONS OF THE GENITALIA

- Genital vascular lesions are uncommon, with **no consensus on classification, cause or treatment**. They comprise **haemangiomas and vascular malformations**, which differ fundamentally:

HAEMANGIOMA VERSUS VASCULAR MALFORMATION

Feature	Haemangioma	Vascular malformation
Site	Skin	Subcutaneous tissue
At birth	Often congenital	Present at birth
Natural history	Postnatal growth, then slow involution	Persists or enlarges — triggered by trauma, sepsis or hormonal change
Subtypes	Strawberry (commonest)	Slow-flow (capillary, lymphatic, venous) or fast-flow (arterial, arteriovenous)

Congenital Haemangiomas

- **About 1% of haemangiomas affect the genitalia**; their origin and the control of growth and involution are controversial.
- **Strawberry haemangiomas are the commonest type** — **proliferations of immature capillaries**. They may grow rapidly for **3–6 months**, then gradually involute; **most need no treatment**.
- **Ulceration requires intervention** to prevent bleeding. The most popular treatment is **short-term oral corticosteroid or propranolol**; **laser therapy** achieves **selective photothermolysis** of superficial vessels; some need excision.

Subcutaneous Haemangiomas

- **Subcutaneous (cavernous) haemangiomas are much rarer and are better classified as vascular malformations**. They appear at birth or later and, unlike cutaneous haemangiomas, **enlarge gradually**.
- They feel like a "**bag of worms**", as a varicocele does, but are **firm and do not decompress when recumbent**.
- Because examination underestimates their extent, **define them with Doppler ultrasound, computed tomography (CT) or MRI**. **Definitive treatment is en bloc resection**; **preoperative angioembolisation reduces size and bleeding**.

Klippel-Trénaunay-Weber Syndrome

- A **triad of cutaneous vascular malformation (most commonly naevus flammeus) with soft-tissue and bone hypertrophy**, present at birth, usually in a **lower limb**, sometimes trunk or face. The lesions **tend to bleed**.
- Genitourinary cutaneous or visceral involvement occurs in **about 30%**; cutaneous genital lesions frequently cause **intractable bleeding**, and **excision is associated with significant blood loss**.

Vascular Malformations

- Penile vascular malformations are congenital but usually **diagnosed in the teens or young adulthood**, as a **faint blue patch or soft blue mass**. **Careful excision** is effective; for **glans lesions the neodymium:yttrium-aluminium-garnet (Nd:YAG) laser** may give a better result.

Concept — *Haemangiomas involute and malformations do not; that single difference in natural history is why haemangiomas are observed unless they ulcerate, whereas malformations are imaged and resected.*

HERNIA AND HYDROCELES

Embryology

- The **processus vaginalis** forms in the **third month of gestation** as the **peritoneum bulges into the inguinal canal just before testicular descent**. After descent it obliterates, and the portion around the testis becomes the **tunica vaginalis**.
- **Only about 18% are closed in term infants**; obliteration continues postnatally, and **failure to obliterate accounts for nearly all inguinoscrotal abnormalities in children**.
- In boys having unilateral hernia repair, a **patent contralateral processus is found in about two-thirds under 2 months and about 40% at 1–2 years**, persisting at similar frequency to 16 years; **incidental patency in older children and adults is about 20%**.

DEFINITIONS — ANATOMY OF THE PROCESSUS VAGINALIS

Entity	Processus vaginalis	Content / communication
Indirect inguinal hernia	Patent beyond the internal ring	Abdominal contents (peritoneal fluid, bowel, omentum, gonads) passing into canal, labia or scrotum
Communicating hydrocele	Patent to the testis	Peritoneal fluid within the tunica vaginalis
Hydrocele of the cord (funicular / encysted)	Patent segment, obliterated distally and proximally	Fluid in the segment; no communication
Scrotal hydrocele	Closed proximally	Fluid in the tunica vaginalis around the testis
Abdominoscrotal hydrocele	Closed	Large scrotal hydrocele extending through the internal ring into the abdomen , no peritoneal communication

- **Figure legend (Fig. 55.27, after Martin et al., 1996):** (A) normal closure — funicular process obliterated, tunica vaginalis around the testis; (B) **communicating hydrocele with complete patency**; (C) **funicular hydrocele with distal closure** — communication with the peritoneal cavity may also produce a hernia; (D) **encysted hydrocele of the cord**.

Concept — Every paediatric hernia and hydrocele is a variation on where, and how completely, the processus vaginalis failed to close; the definitions differ only in the level and extent of patency.

Epidemiology and Pathogenesis

Inguinal Hernia and Communicating Hydrocele

- **Inguinal hernias occur in 1–5% of children**, are **5–10 times commoner in boys**, and are **much commoner in premature infants** (about 13% of babies born before 32 weeks and nearly 30% of those under 1 kg).
- **Right-sided predominance (3:1)** is attributed to **later descent of the right testis**. **Risk factors for bilaterality: female sex, prematurity, age under 1 year and cryptorchidism**. Breastfeeding may be protective.
- Hernias present at any age, but the **mean age is 3–4 years**, with **nearly one-third before 6 months**. **Concomitant hydroceles are frequent**, most scrotal, some of the cord.
- **Most new hydroceles after birth and before puberty are communicating**, although a proportion of apparently non-communicating and cord hydroceles resolve spontaneously. **The processus is found completely obliterated in only a minority at hydrocele repair**.

Noncommunicating Hydroceles

- Hydroceles that **resolve in infancy, or appear at or after puberty, are usually non-communicating**. They arise from **delayed absorption or abnormal fluid dynamics** within the tunica vaginalis or, less commonly, in response to **tumour, trauma or inflammation**.

- **Simple scrotal hydroceles occur in at least 5% of male neonates, are typically bilateral, and resolve by reabsorption.**

Genetics and Associated Conditions

- There is a **genetic predisposition**: relatives of probands are at increased risk, highest in **first-degree relatives and especially sisters of affected girls**; transmission appears **autosomal dominant with reduced penetrance**.
- Associations: **bladder exstrophy, epididymal anomalies, connective tissue disorders, cystic fibrosis and posterior urethral valves**; inguinal hernia is a feature of **more than 200 syndromes**. Findings at repair may include **polyorchidism, transverse testicular ectopia ($\pm$ persistent Müllerian ducts) and splenogonadal fusion**. **Abnormal epididymal–testicular attachment correlates with the degree of processus closure**. **Ascites or a ventriculoperitoneal shunt increases hernia risk**.

Concept — *Hernia risk tracks anything that keeps the processus open (prematurity, later right-sided descent, cryptorchidism) or raises intra-abdominal fluid and pressure (ascites, shunts).*

Diagnosis

Signs and Symptoms

- Hernias and communicating hydroceles present as a **painless groin bulge**, sometimes extending along the cord to the scrotum, often **only with raised intra-abdominal pressure** (crying, straining) and reducing when supine. **An intermittent bulge distinguishes a reducible hernia or communicating hydrocele from a scrotal or cord hydrocele.**
- **Incarceration**: an irritable or inconsolable child with a **persistent or larger bulge that does not reduce**, poor feeding, and signs of **bowel obstruction** (distension, vomiting, no flatus or stool).
- A **scrotal hydrocele** may appear chronically or acutely after inflammation, infection or trauma; its size is stable or decreases. A **cord hydrocele** is painless, variable in size, and may be **mistaken for the testis** because of its oval shape.

Physical Examination

- Examine **standing if age-appropriate**, otherwise supine; inspect from the lower abdominal crease along the canal to the scrotum, watching a crying child's bulge appear and subside. Provoke it by **restraining an infant to induce crying**, or by **jumping, coughing, laughing, blowing bubbles or balloons** in older children. **If no bulge is elicited, parents' photographs are diagnostically reliable.**
- **Palpate craniocaudally from superolateral to the pubic tubercle down to the scrotum** to define proximal and distal extent: hernias and communicating hydroceles **start at the internal ring**. The **silk-stocking sign** — the sensation of rubbing silk when the cord is rolled over the pubic tubercle — indicates **thickened cord structures**.
- A **cord hydrocele** can mimic a testis or large epididymal cyst, but **normal cord is palpable above and below it and the testis is felt separately**. A scrotal hydrocele may show a **blue hue**; if it reaches or crosses the internal ring (abdominoscrotal), **abdominal examination reveals a ballotable mass**.
- **Hydrocele fluid transilluminates — but so may neonatal bowel**. The testis is palpable in a soft hydrocele but may be impalpable in a tense one.

Radiologic Imaging

- **Imaging is not usually needed**. Ultrasound helps in the **obese child** (an elongated echolucent track from the groin along the cord), to **confirm bowel or omentum in an incarcerated hernia**, to distinguish a **tense**

cord hydrocele from a solid mass or epididymal cyst, and to identify an impalpable testis within a hydrocele.

Concept — *Fluctuation is the key discriminator: a bulge that comes and goes means a patent processus (hernia or communicating hydrocele) and therefore surgery, whereas a fixed collection means a closed processus and permits observation.*

Surgical Repair

Inguinal Hernia

- **Inguinal hernias need repair shortly after diagnosis because of the risk of complications.** Easily reducible hernias and communicating hydroceles are repaired as outpatients within a few weeks; **difficult reduction → more urgent repair; an irreducible hernia needs immediate exploration.** Counsel parents on the signs of incarceration.
- **Premature infants** can be repaired before discharge, but repair may be delayed in **extremely low-birth-weight infants (under 1500 g)** or those with **congenital heart disease, pulmonary disease, sepsis or metabolic disease** because of anaesthetic risk; some advise waiting until **50–52 weeks postconception.**

Hydroceles

- **Infant hydroceles can resolve spontaneously — most by 1 year, some by 2 years.** Persistence, enlargement or new fluctuation (implying a communicating hydrocele/hernia) **is an indication for surgery.** Cord hydroceles rarely resolve but rarely need urgent surgery, and can be corrected after 1 year.

Standard Inguinal Hydrocele/Hernia Repair

- **High ligation of the sac at the internal ring** has a high success rate, low morbidity, little pain and good cosmesis.
- Sequence: **skin-crease incision in Langer's lines superolateral to the pubic tubercle** → incise **Scarpa's fascia** → clear the external oblique laterally and caudally to the external ring → **open the external oblique in the line of its fibres, sparing the ilioinguinal nerve** (in young infants the rings nearly coincide, so the external oblique need not be opened — **Mitchell-Banks technique**) → separate **cremaster fibres perpendicular to the cord** → find the sac **anteromedially** → separate it atraumatically and **divide between clamps** → dissect the proximal limb to the internal ring and **ligate there** to prevent recurrence (**open the sac first if contents are present or suspected, or if it is large or thick**, to reduce bowel, omentum or a sliding component) → **tighten a widened internal ring medially without compressing the cord** → leave a short distal sac; excise the anterior wall of a long one → excise redundant tunica vaginalis without encroaching on the cord, and remove appendages.
- **Peripubertal hydrocele:** use an **inguinal approach if the hydrocele is clinically communicating; otherwise a trans-scrotal repair** as in adults, converting to inguinal if a proximal communication is found. Hydroceles are **non-communicating in the large majority of boys over 10–12 years**, which supports a scrotal approach from that age.

Complications

- **Early bleeding and infection are rare.**

RECURRENCE AFTER INGUINAL HERNIA REPAIR

Setting	Recurrence
Uncomplicated open repair	0.5–1%
Premature infants	2%

Setting	Recurrence
After incarcerated hernia	3–6%

- **Half of recurrences appear within 1 year and more than three-quarters within 2 years.** Causes: **failure to identify or ligate the sac; a torn sac leaving a peritoneal strip on the cord; damage to the floor; or a missed direct hernia.** (One cohort in which high ligation was deliberately omitted to protect adjacent structures had no recurrences.) Recurrent hernias are repaired inguinally, **ligating an indirect sac at the internal ring and/or repairing the floor for a direct hernia.**
- **Genital complications:** iatrogenic cryptorchidism is avoidable by confirming testicular position at the end of the repair; an **incompletely descended or retractile testis should be orchidopexied at the herniorrhaphy.** Testicular atrophy is rare in large series but more frequent in others and **after irreducible hernias,** from **compression of gonadal vessels or surgical vascular injury;** young infants (under 2–3 months) have **infarction rates of about 30%.** True vas deferens is found in specimens in a **small fraction of a percent** (embryonal remnants are commoner); **vasal injury is likelier in infants and incarceration,** and non-transecting injury may cause **obstruction and infertility later.**
- **Persistent hydrocele after decompression at hernia repair is rare → observe for at least 1 year → then scrotal hydrocele repair, or re-exploration if fluctuation suggests recurrent hernia.**

Scrotal Approach to Inguinal Hernia Repair

- Paralleling scrotal orchidopexy, a **scrotal approach** (a **median raphe incision for bilateral hernias**) ligates the processus at the **external ring.** Early data show **similar success and morbidity with a faster operation and cosmetic benefit,** but the **level of ligation and the effect of age need long-term follow-up.**

Laparoscopic Inguinal Hernia Repair

- Two techniques: **intraperitoneal peritoneal closure of the defect** and an **extraperitoneal approach** (of growing interest). Success and complication rates **now approach or equal open repair,** the convergence explained by **experience and technical refinement.**
- Pooled comparisons against open repair: earlier meta-analysis — a trend to **higher recurrence,** longer unilateral and shorter bilateral operating times, and **significantly fewer contralateral metachronous hernias;** later — **similar recurrence, fewer ascended testes and metachronous hernias, but more wound infection;** the most recent — **shorter operative times for unilateral and bilateral hernias (longer for unilateral hernias in girls), similar recurrence and fewer complications.**
- **Surgeon preference remains open repair:** in a European survey, mostly of senior and experienced laparoscopic surgeons, **most favoured open repair for its lower recurrence and lower risk of injury to adjacent structures;** the minority preferring laparoscopy cited **fewer metachronous contralateral hernias, cosmesis, ease, less vas/vessel injury and less pain, and few would use it for incarceration.**
- **Miniaturisation:** two-port and single-port techniques give similar outcomes; in **infants of 5 kg or less** microlaparoscopic repair (2-mm instruments) had low recurrence (**higher with American Society of Anesthesiologists (ASA) score ≥ 3**), no atrophy, but some **high testes needing surgery.** Division of the sac distal to the ring with a **non-absorbable purse-string** works in infants under 1 year, whose recurrence rate is **equal to or lower than older children's.**
- **Laparoscopy for incarceration** has three stated advantages: **avoiding the oedematous cord tissue, reducing bowel under direct vision, and inspecting the incarcerated organ at the end.**

APPROACHES TO PAEDIATRIC INGUINAL HERNIA REPAIR

Feature	Open inguinal	Scrotal	Laparoscopic
Level of ligation	Internal ring (high ligation)	External ring	Internal ring (peritoneal closure)

Feature	Open inguinal	Scrotal	Laparoscopic
Evidence	Longest record; highest success	Early, short follow-up	Now approaching open
Advantages	Low recurrence, familiarity	Faster, cosmetic	Inspects and treats contralateral ring; fewer metachronous hernias; incarceration under vision
Disadvantages	Contralateral side unseen	Ligation level and age effects unresolved	Wound infection; learning curve; high testes in infants

Concept — *Ligation of the patent processus at the internal ring is the whole operation, whatever the access; the approaches differ only in trade-offs — laparoscopy wins by seeing and closing the other side, open repair by its recurrence record.*

Assessment of the Contralateral Internal Ring

- **Whether to assess the contralateral processus in unilateral hernia remains controversial.** The aim is to **avoid metachronous hernia and its risks and costs.** Routine contralateral exploration gave way to **selective exploration** in high-risk groups (young age, prematurity, sex, associated disease); surveys show many surgeons still explore premature infants and those under 2 years.
- **Laparoscopy — via the umbilical port when a laparoscopic repair is planned, or transinguinally through the sac at open repair — gives a rapid, direct and accurate view of the contralateral ring,** with sensitivity and specificity **above 99%** on meta-analysis. **Bubbles of air escaping through the ring indicate patency.**
- Contralateral patency is **inversely related to age** — about 40% in infants, 30% in toddlers and 20% after 10 years, **similar to the 20% found at adult autopsy** — and is found more often at open exploration than at laparoscopy.
- **Metachronous hernia is uncommon: about 7% when contralateral status is unknown (90% of them within 5 years),** and lower still when laparoscopy shows a closed ring; **closing a patent contralateral processus significantly reduces metachronous hernia.**
- These data are indirect: **the natural history of a known open ring remains unknown** until a prospective study of such rings is done.

Concept — *A patent processus is far commoner than a clinical hernia; that gap is the entire argument against routine contralateral closure — many more rings are open than will ever herniate.*

Abdominoscrotal Hydrocele

- **Uncommon (about 1% of hydroceles) and bilateral in about 30%.** A **ballotable scrotal mass extending into the abdomen,** palpable there when large; **ultrasound defines the proximal extent.** It presents in infancy, either as such or as a scrotal hydrocele that **enlarges, improves or resolves.** Associated: **cryptorchidism, contralateral hernia or hydrocele, vanishing testis.**
- **Mechanism:** a scrotal hydrocele that **enlarges into the retroperitoneal or properitoneal space after closure of the processus;** the fluid is **exudative,** and rising size and pressure may cause **lymphatic obstruction and progression.** Massive extension can cause **hydroureteronephrosis, lower-limb oedema or appendicitis.** **Dysmorphic elongation of the testis** occurs but is **usually reversible.**
- **Surgery:** traditionally **inguinal,** dissecting the sac from its abdominal attachments and mobilising it distally, **± orchidopexy to prevent iatrogenic cryptorchidism; aspirating the scrotal component** eases proximal dissection. Alternatives: **midline abdominal approach for large bilateral cases,** or **laparoscopic decompression followed by inguinal excision.** A strip of sac may be left on the cord to protect cord and vas. **Tightening the internal ring is unnecessary — the processus is invariably closed.** A **primary scrotal approach** (drainage, excision, extensive plication with limited dissection) had **no recurrences and fewer complications than inguinal repair** in a comparative series.

KEY POINTS: HERNIAS AND HYDROCELES

- Inguinal hernias occur in **1–5% of children**, commoner in **boys and premature infants**; **right-sided (3:1)** from later right descent; **bilaterality** risk with **female sex, prematurity, age under 1 year and cryptorchidism**.
- An **intermittent bulge distinguishes reducible hernia and communicating hydrocele** from scrotal or cord hydrocele; **home photographs are reliable**.
- **Repair hernias soon after diagnosis; observe hydroceles**, which often resolve.
- Repair may be inguinal, scrotal or laparoscopic; **the inguinal approach currently has the highest success**.
- Recurrence (**0.5–1%**) follows **unidentified or unligated sac, torn sac, damaged floor or missed direct hernia**.
- **Diagnostic laparoscopy gives rapid, direct and accurate inspection of the contralateral ring**.

Concept — *The abdominoscrotal hydrocele is a closed-processus hydrocele that has expanded upwards under pressure: being non-communicating, it needs excision or drainage rather than high ligation, and its mass effect explains its complications.*

ACUTE SCROTUM

- **Acute scrotum is new-onset pain, swelling and/or tenderness of the intrascrotal contents.** The differential is limited but signs overlap; some clinical features are reliable, and scrotal imaging helps.
- **Frequency: torsion of the appendix testis is commonest (40–60%), then spermatic cord torsion (20–30%, excluding neonates), epididymitis (5–15%), and other or no pathology (about 10%).**
- **Age pattern:** appendage torsion is typical **after infancy and before puberty**; epididymitis and cord torsion peak in the **perinatal and pubertal periods**.
- **Appendage torsion and epididymitis are managed conservatively with limited consequence; testicular torsion needs prompt exploration** because the gonad is at risk of ischaemic loss, especially with delay in presentation, evaluation or treatment.
- **Delay in presentation** stems from embarrassment or denial and from **poor public awareness**: adolescents and parents are often unaware of torsion, adolescents fear raising false alarms and families fear burdening services; most parents have never discussed it with their child, and few learned of it from their paediatrician. A short **electronic module for clinicians** improved confidence, and a **public educational video** was produced.

BOX 55.2 DIFFERENTIAL DIAGNOSIS OF PAEDIATRIC AND ADOLESCENT ACUTE SCROTAL PAIN

Category	Entities
Appendage torsion	Appendix testis; other appendages (epididymis, paradidymis, vas aberrans)
Spermatic cord torsion	Intravaginal (acute or intermittent); extravaginal
Epididymitis	Infectious — UTI, sexually transmitted disease, viral; sterile or traumatic
Scrotal oedema or erythema	Diaper dermatitis, insect bite, other skin lesions; idiopathic scrotal oedema
Orchitis	With epididymitis ± abscess; vasculitis (e.g., Henoch–Schönlein purpura); viral (mumps)
Trauma	Haematocoele, scrotal contusion, testicular rupture
Hernia or hydrocele	Inguinal hernia ± incarceration; communicating hydrocele; encysted hydrocele ± torsion; with acute abdominal pathology (appendicitis, peritonitis)

Category	Entities
Varicocele	—
Intrascrotal mass	Cystic dysplasia or tumour of testis; epididymal cyst, spermatocele or tumour; other paratesticular tumours
Musculoskeletal	Inguinal tendonitis, muscle strain
Referred pain	Ureteral calculus or anomaly

- **What the box is for:** the differential is broad, but only **cord torsion** is time-critical; note the traps — **referred pain (ureteral calculus)** and **abdominal pathology (appendicitis)** can present in the scrotum, and **Henoch-Schönlein purpura** can mimic or cause torsion.

THE THREE COMMONEST CAUSES COMPARED

Feature	Spermatic cord torsion	Appendage torsion	Epididymitis
Peak age	Perinatal; 12–16 years	7–12 years	Infancy and puberty
Onset	Acute, severe ; may follow earlier episodes	Sudden or insidious; intermittent, worse with activity	Insidious (may be rapid)
Nausea/vomiting	Common	Uncommon	Rare
Cremasteric reflex	Usually absent	Symmetrical, present	Present
Signature sign	High-riding, horizontal testis; generalised tenderness	Blue dot sign ; tender upper-pole nodule	Epididymal enlargement; pyuria in a minority
CDUS	Reduced/absent flow, heterogeneity; cord twist ("whirlpool"/"snail")	Avascular upper-pole lesion; epididymal hyperperfusion	Increased epididymal (± testicular) size and flow
Management	Emergency exploration	Conservative	Conservative ± antibiotics

Concept — Only one diagnosis in the acute scrotum is time-critical, so the entire clinical and imaging strategy exists to rule spermatic cord torsion in or out quickly; everything else can safely wait for it to be excluded.

Spermatic Cord Torsion

Acute Intravaginal Spermatic Cord Torsion

Predisposing factors.

- Intravaginal torsion is attributed to **excess testicular mobility in a "bell-clapper" deformity, in which the tunica vaginalis attaches abnormally high on the cord**. The deformity is found in **about 12% of males at autopsy**, yet torsion is far rarer (**about 8.6 per 100,000 males aged 10–19 per year**).
- There is a **familial predisposition** (mode of transmission unknown), including multigenerational families.
- The trigger is unknown; suggested are **cold or a change in temperature activating the cremasteric reflex** (even in tropical countries) and **rapid pubertal testicular growth** — yet torsion also occurs at rest or during sleep.
- **Cryptorchid testes are at increased risk** and are hard to assess because of their position. **Torsion after orchidopexy** is attributed to **failure of the fixing suture, absorbable or non-absorbable**.

Clinical presentation.

- Torsion can occur at any age, but **most cases are after 10 years, peaking at 12–16 years**. Its prevalence is **1 in 4000**, with **left-sided predominance and rare bilaterality**.
- Classically there is **acute, severe scrotal pain** at rest (even during sleep), with activity or after trauma, sometimes with **previous episodes**. Pain may instead be **milder, less acute, absent, or felt in the groin or**

abdomen. Nausea and vomiting occur in 10–60%. Oedema and erythema depend on duration and degree; **dysuria and fever are uncommon.**

- **Commonest findings: generalised testicular tenderness, abnormal testicular orientation (high-riding from a shortened cord; horizontal lie) and an absent cremasteric reflex.**
- The **cremasteric (genitofemoral) reflex**, normally present **after age 2 years**, is elicited by stroking the inner thigh. It is absent in torsion in some series but **intact in up to 10% of proven torsions** in others: **its presence correlates with intact flow but does not exclude torsion**, particularly when the presentation is otherwise suggestive.
- Anterior epididymis, cord thickening, testicular induration, loss of the testis–epididymis boundary, oedema and erythema may be present, but **landmarks disappear and examination becomes less reliable as duration increases.**
- **Clinical tools:** on multivariate analysis **absent cremasteric reflex, nausea/vomiting and scrotal skin changes predict torsion.** A decision tool found **no torsion in patients with acute pain (under 72 hours) who had a normal testicular lie, no nausea or vomiting, and age 0–10 years** (complete sensitivity and negative predictive value). The **TWIST score** stratifies risk using these factors and has been validated, and a **standardised scoring pathway shortened the time from emergency department or ultrasound to theatre.**

Diagnostic studies.

- **Urinalysis has limited value in torsion** but identifies **pyuria/bacteriuria (epididymitis) or haematuria (calculus).**
- Before rapid imaging, **immediate exploration was routine.** **Radionuclide imaging** (about 90% sensitive and specific) is slow, not readily available and uses ionising radiation — **now rarely used.** **Ultrasound** is rapid, available and safe, showing architecture, parenchymal flow and details such as hydrocele and wall thickening.
- **CDUS findings of torsion: reduced or absent colour flow or waveforms, and parenchymal heterogeneity compared with the other testis.** **Heterogeneous echotexture indicates necrosis**, whereas **homogeneous echotexture predicts a lower risk of orchidectomy.**
- **Absent intratesticular flow is highly specific for torsion** (classic series: sensitive, 100% specific, 97% accurate). In a blinded prospective study CDUS had **100% negative predictive value**, leading its authors to conclude **routine exploration is unnecessary when perfusion is normal — but other series report sensitivity of only 63–90%**, because newer equipment detects residual flow in partial torsion and because the test is operator-dependent.
- **Supportive Doppler signs:** subtle waveform abnormalities — **increased or decreased amplitude and reversal of diastolic flow** — and an **enlarged, altered-echogenicity epididymis with absent, reduced or occasionally increased flow.**
- **High-resolution ultrasonography (HRUS, 10–20 MHz) of the whole cord:** the "snail" and "whirlpool" signs of a twisted cord are reliable; a cord twist was seen in almost all surgically proven torsions, and a linear cord was highly specific against torsion — **but partial or intermittent torsion can show normal flow and no visible twist.**
- **Emerging modalities:** **contrast-enhanced pulse-inversion ultrasonography (CEUS)** — superior to CDUS for indeterminate testicular lesions; **infrared thermography** (scrotal cooling after torsion in animals); **trans-scrotal near-infrared spectroscopy** (discriminated torsion in selected children); **MRI — dynamic contrast-enhanced (requires gadolinium) or diffusion-weighted, showing a lower apparent diffusion coefficient (ADC) without contrast.**

- **Figure legends (Figs 55.32–55.33):** in **intermittent torsion**, CDUS may show preserved arterial flow early and **loss of flow with heterogeneity when pain worsens**; acute torsion shows **reduced arterial flow**; **prolonged torsion** shows a **heterogeneous testis without arterial or venous flow** and a **hyperechoic parenchymal ring**. The **snail sign** is a **hyperechoic central "body" with blood coiling around it** in the distal cord.

Management and Surgical Treatment

- **Testicular torsion is a true surgical emergency because viability is inversely related to duration.**

TIME FROM ONSET OF PAIN AND TESTICULAR OUTCOME (POOLED PUBLISHED SERIES)

Hours from onset	Risk of orchidectomy	Delayed atrophy after orchidopexy
0–6	~5%	< 10% (under 12 h)
7–12	~20%	
13–18	~40%	~40% (12–24 h)
19–24	~60%	
> 24	~80%	~75%
> 48	~90%	

- **Reading the table:** loss rises steeply after 6 hours, and **even a testis saved at surgery may atrophy later**. The **degree of twist** explains some of the variability — incomplete occlusion slows loss, or twist may multiply the effect of time; **lost testes have a greater median twist (about 540° versus 360°), though the ranges (180–1080°) overlap**. **In-hospital delay — time from emergency department to detorsion — independently lowers salvage.**
- **Orchidectomy follows exploration in 30–70% of large series and 32–42% of database reviews. Partial atrophy can follow detorsion even after short durations of about 4 hours, and atrophy (to less than half the contralateral volume) is common even when the testis was judged viable, predicted by pain over 1 day and sonographic heterogeneity.**
- **Manual detorsion** may relieve symptoms and allow **delayed orchidopexy**, but **may be incomplete: about one-third of cords are twisted in the opposite direction to that assumed**, and manual detorsion then fails.
- **Two strategies for the acute scrotum:**

EXPLORATION STRATEGIES

Strategy	Basis	Cost
Explore almost all	Avoid delay; confirm absence of torsion	Many unnecessary operations
Selective exploration	History, examination and CDUS	Testis loss from atypical presentation or false-negative imaging

- **When findings support or raise suspicion of cord torsion, emergent exploration is indicated and must not be delayed.**
- **Operative sequence: hemiscrotal transverse (dartos pouch) or midline raphe incision → affected side first → deliver the testis, open the tunica vaginalis, note colour, number of rotations and tunica anatomy → detorse, wrap in warm moist gauze and observe → meanwhile fix the contralateral testis with non-absorbable suture to prevent metachronous torsion → re-examine the affected testis and make the largely subjective decision between orchidopexy and orchidectomy (a Doppler probe, or incising the tunica albuginea and watching for bleeding, may document flow, but neither is validated) → a retained testis is fixed in a dartos pouch or directly to the dartos with non-absorbable suture.**

- **Compartment-syndrome concept:** improved appearance and lower intraparenchymal pressure after detorsion and **tunica albuginea incision** suggest a compartment syndrome contributes. A **vascularised tunica vaginalis flap** placed in the albugineal defect keeps pressure low; some testes judged non-salvageable recovered, but long-term outcome is unknown and animal studies show no histological advantage.
- **Other risk factors for orchidectomy:** **young age, Black race, Medicaid or no insurance, and interhospital transfer.** Transfer delays treatment; it did not change pooled orchidectomy rates, but **harmed salvage in those presenting within 24 hours of onset.**
- **Orchidectomy:** divide the cord into segments, each **suture-ligated.** A **prosthesis may be offered after complete healing or later in puberty;** concurrent placement has been reported.

Prognosis

- The effect on **fertility is poorly understood,** but **subtle semen abnormalities are common.** **Sperm density is often normal, and correlates with shorter torsion and less atrophy.** The inverse relation of duration to semen quality, and contralateral biopsy data, suggest **global testicular dysfunction after torsion.**
- The **autoimmune (antisperm antibody) hypothesis has been dispelled;** animal and human data support **ischaemia-reperfusion injury.** Years after torsion, **FSH, LH and testosterone are normal but inhibin B is reduced** in both detorsed and orchidectomised patients, with no difference between them. More long-term data are needed, including on adjunctive treatments.

Intermittent Intravaginal Spermatic Cord Torsion

- **Self-limited episodes of acute scrotal pain precede acute torsion in 30–50% of patients.** Episodes, single or multiple, begin and resolve abruptly over **minutes to hours,** and **their frequency correlates with eventual torsion and testicular loss.**
- Nausea, vomiting or swelling may be absent. Findings depend on whether the testis is twisted at the time; **a normal vertical lie is commonest** (horizontal lie possible), and in series **the cremasteric reflex was usually present and CDUS flow usually normal.** A **whirlpool sign, an abnormal boggy cord, or a pseudomass below the twist** may indicate intermittent torsion.
- **Diagnosis needs a high index of suspicion** unless the testis is seen to untwist, or CDUS shows **absent/reduced flow before and normal/increased flow after marked relief of symptoms.**
- **Once confirmed or highly suspected, elective bilateral orchidopexy is indicated** to avert torsion and loss. Patients and parents should know that **absolute confirmation may be impossible and symptoms may persist.**

Extravaginal Spermatic Cord Torsion (Perinatal Testicular Torsion)

- **Perinatal torsion** applies to infants whether the event is prenatal (hours to months before birth), during delivery or postpartum. **The whole cord twists before the tunica vaginalis and dartos are fixed within the scrotum (extravaginal).**
- It usually occurs **well before delivery,** producing a **"vanishing" testis or a haemosiderin-containing nubbin** in the scrotum or, less often, the canal. Loss near delivery gives a **hard, painless testis fixed to erythematous or dark scrotal skin, ± oedema;** a **coexisting hydrocele** complicates examination.
- **Incidence about 6.1 per 100,000 births;** familial cases occur. **High birth weight and difficult delivery** are suggested but unconfirmed. In a minority, symptoms follow a documented normal examination. **Rarely, neonatal intravaginal torsion or infarction without torsion** occurs.
- **Bilateral torsion occurs in a minority (5–22%),** synchronous or **metachronous — even when the first event was prenatal.**

- **Imaging is of questionable usefulness.** Prenatal ultrasound may show torsion; postnatal ultrasound may show **heterogeneity, calcification and absent flow**, but **CDUS can show flow in torsed testes**, and ultrasound's ability to distinguish **tumour from torsion** has also been challenged.
- **There is no consensus on treatment:**

PERINATAL TORSION — ELECTIVE VERSUS IMMEDIATE EXPLORATION

Elective exploration	Immediate exploration
Testis rarely salvageable	Offers possible partial or complete salvage
Metachronous contralateral torsion is rare	Unexpected contralateral torsion or atrophy found at exploration
Anaesthetic risk in the neonate	—

- In a UK and Ireland survey, **few used Doppler to guide management or exclude tumour; most performed ipsilateral orchidectomy and contralateral orchidopexy but few operated emergently**, because a viable testis was rarely found; some **avoided contralateral orchidopexy for fear of iatrogenic injury**.
- **If torsion is suspected after a documented normal postnatal examination, explore promptly as for intravaginal torsion.** Some use a **scrotal** approach; others an **inguinal** approach to ligate a patent processus and **avoid trans-scrotal violation if a tumour is found**.

INTRAVAGINAL VERSUS EXTRAVAGINAL TORSION

Feature	Intravaginal	Extravaginal (perinatal)
Anatomy	Testis twists within the tunica vaginalis — bell-clapper	Whole cord and tunica twist before scrotal fixation
Age	Peak 12–16 years	Prenatal to neonatal
Presentation	Acute pain, high horizontal testis, absent reflex	Painless hard testis, discoloured fixed skin , or vanished testis/nubbin
Salvage	Time-dependent — often salvageable if early	Rarely salvageable
Urgency	Emergency exploration	Debated ; urgent if it follows a normal postnatal examination
Contralateral	Fix with non-absorbable suture	Usually fixed; some avoid

Concept — *Timing relative to testicular fixation defines the type and the urgency: before the tunica fixes, the whole cord twists and the testis is usually dead at birth, so urgency is debated; after puberty the bell-clapper testis twists within the tunica, and every hour counts.*

Torsion of the Appendix Testis and Epididymis

- **Appendage torsion is the commonest cause of acute scrotum in prepubertal children.** The appendages are functionless embryological vestiges:

TESTICULAR APPENDAGES

Appendage	Origin	Present in	Site
Appendix testis	Müllerian duct	~76–83% of testes	Cranial pole, or groove between testis and epididymis
Appendix epididymis	Wolffian duct	~22–28%	Along the caput

- Either may be **sessile or pedunculated**; the sessile type may be commoner, but the **pedunculated type is more prone to torsion**. The cause may relate to anatomy, trauma or **prepubertal enlargement**.
- **Peak age 7–12 years (mean 8–9)**, though any age is possible. Onset may be sudden or insidious, pain variable and intermittent, **worse during or after activity**; signs depend on inflammation and duration.

- **Early: the "blue dot sign" — the ischaemic appendage seen through stretched skin at the upper pole —** present inconsistently; a **false-positive blue dot in cord torsion has been reported**. Other early signs: a **tender nodule above the testis, limited testicular tenderness and symmetrical cremasteric reflexes**. Later, swelling, oedema and erythema make it **indistinguishable from torsion or epididymitis**.
- **CDUS rarely shows the abnormal appendage but commonly shows epididymal hyperperfusion**. The normal appendix has no flow; a torsed one appears as a **round or oval avascular lesion with heterogeneous echotexture, posterior enhancement and surrounding hyperaemia near the upper pole**. It is **hypoechoic within 24 hours**, and variable (hypo-, iso- or hyperechoic) thereafter.
- **Because appendage torsion is self-limiting, surgery is rarely indicated**. Treat with **ice packs, oral anti-inflammatory agents, scrotal support and restricted activity**. Explore only when cord torsion cannot be excluded, or rarely for prolonged severe or recurrent pain.

Epididymitis

- A broad category, **infectious or non-infectious**; its incidence is poorly defined, partly because a **torsed appendage may be misdiagnosed as epididymitis**. Prevalence peaks in **infancy and at puberty**; in a long clinic series most cases were pubertal (11–14 years), few under 2 years, and **about a quarter had a second episode**.
- **Onset is classically more insidious** than cord or appendage torsion but may be rapid. **Fever and dysuria are variable; nausea is rare**. Look for a history of **UTI, urethral discharge, sexual activity, intermittent catheterisation, dysfunctional voiding, urethral abnormality or ejaculatory duct anomaly**.
- Examination ranges from **localised epididymal enlargement and tenderness to a massively swollen, erythematous hemiscrotum without landmarks**. The **cremasteric reflex should be intact**. **Pyuria and/or bacteriuria in 20–40%**. CDUS shows **increased epididymal size and flow**, sometimes testicular too.
- **Management aims to relieve inflammation and treat infection: ice, non-steroidal anti-inflammatory drugs (NSAIDs), scrotal elevation and rest**. Without UTI, symptoms resolve without antibiotics. With pyuria, give **broad-spectrum antibiotics with Gram-negative cover**, intravenously with admission if needed; **consider sexually transmitted disease treatment in adolescents**.
- **Imaging is now selective**. Early series found many anatomical anomalies (urethral anomalies, ureteral ectopia) in prepubertal boys; later series found a low risk except in **infants**, with anomalies commoner on VCUG in **culture-positive** patients. Some advise renal ultrasound only, or imaging only for recurrence. **In a prepubertal boy with a positive urine culture, renal ultrasound and VCUG are indicated**. Findings such as pyelectasis or low-grade VUR may be incidental, but **urethral or ureteral obstruction and ejaculatory duct or vasal anomalies** matter.
- **Recurrent epididymitis: endoscopy may reveal ejaculatory duct anomalies missed on VCUG**; recurrent epididymo-orchitis may need **excision of a utricle or ejaculatory duct cyst with vasovasostomy or vasectomy**.

Other Causes of Acute Scrotal Pain

OTHER CAUSES OF ACUTE SCROTAL PAIN

Condition	Features	Management
Idiopathic scrotal oedema	Prepubertal; uni- or bilateral swelling with minimal or no pain ; spreading erythema/oedema to groin or perineum; inguinal lymphadenopathy, leucocytosis, eosinophilia (suggesting allergy) ; isolated scrotal-wall tenderness and thickening; "fountain sign" — hypervascular thickened scrotal wall on transverse Doppler with both testes in view — highly suggestive	Self-limiting ; antihistamines at most

Condition	Features	Management
Henoch–Schönlein purpura	Systemic vasculitis of skin, joints, gut and kidney; scrotum involved in 2–38% — tenderness, oedema, erythema, testicular haematoma, torsion , infarction, cord thrombosis, epididymitis; purpuric rash (buttocks, perineum, legs) may not precede scrotal signs	Urgent exploration if torsion is suggested
Cystic dysplasia of the rete testis	Benign; first described in 1973 ; commonest around 6 years ; acute swelling and pain (rarely chronic pain from an associated ureteral anomaly); ipsilateral renal dysplasia, agenesis or ureteral anomaly in most (Wolffian maldevelopment); ultrasound: multiple small (2–8 mm) central cysts with compressed normal parenchyma; MRI helpful	Conservative enucleation (may recur) or observation — spontaneous regression reported
Testicular rupture	Trauma; pain, swelling, haematocele; CDUS sensitive for fracture	Surgery, or conservative management , reserving surgery for expanding haematoma, symptomatic hydrocele, intractable pain, absent flow or abscess (observation achieved salvage in a small series)

KEY POINTS: ACUTE SCROTAL PAIN

- **Delay in presentation** stems from embarrassment or denial and from **lack of public awareness** of torsion.
- A **reliable history and targeted examination** give the diagnosis in most cases; **ultrasound confirms**.
- **Ultrasound features of cord torsion: absent flow, a "snail sign" and parenchymal heterogeneity.**
- **Salvage is inversely related to duration** and partly to degree of torsion — **when torsion is highly suspected, do not delay surgery for imaging.**
- **Perinatal torsion identified in the nursery is rarely salvageable**; manage emergently **when the examination changes.**
- **Appendage torsion is the commonest prepubertal cause**; the **blue dot sign** may be seen; it is **self-limiting**, needing supportive care and analgesia.

VARICOCELE

- **Varicocele** — abnormal dilatation and tortuosity of the internal spermatic veins of the pampiniform plexus — is common in adolescents and may contribute significantly to adult subfertility. Its effect on paternity is less clear: most men with varicocele in population studies have fathered children.

Epidemiology and Pathogenesis

- **Prevalence is 8–16% in adolescents, similar to about 15% in adults**; since varicoceles **very rarely resolve** and rarely appear in older adults, the adolescent and adult groups are presumed to be the **same population**.
- **Most appear after 10 years, progress through puberty, and peak at Tanner stage 3.** Historically left-sided, **concurrent right-sided varicoceles are increasingly recognised**: bilateral repairs were done in a third of one young series, and CDUS shows subclinical right or bilateral reflux in many boys with left varicoceles — of unclear clinical significance. In adults, **bilaterality is 15–50%**.
- **A palpable isolated right varicocele warrants abdominal and pelvic ultrasound for a mass obstructing drainage to the vena cava.**

- **Pathophysiology is multifactorial:** the primary factors are **raised pressure in the left renal vein, collateral venous anastomoses, and incompetent valves at the junction of the left internal spermatic vein with the left renal vein.** Progression may relate to **continuous or spontaneous (not Valsalva-induced) reflux.** The "**nutcracker phenomenon**" — **compression of the left renal vein between the aorta and superior mesenteric artery** — explains some cases.
- A **generalised venous abnormality** is suggested by abnormal prostatic plexus flow in bilateral varicoceles and by associations with **saphenofemoral incompetence and varicose veins.** Venography shows that **adolescents more often have congenital abnormalities (fewer competent valves, nutcracker phenomenon),** whereas **adult reflux arises through collateral pathways.**
- **Genetics:** varicocele is **4–8 times commoner in first-degree relatives,** especially **brothers.**
- A **tall, thin habitus (low body mass index, BMI)** is associated internationally — perhaps from a **longer spermatic vein and higher hydrostatic pressure,** or simply **easier detection on examination.**

Diagnosis and Classification

- Most are found **incidentally** by a primary care practitioner or on self-examination; **pain is an uncommon presentation (2–11%).**
- **Examine supine and standing;** inspect for visible swelling, then palpate the cord at rest and with **Valsalva.**

CLINICAL GRADING OF VARICOCELE

Grade	Definition
0 (subclinical)	Not palpable; seen only on CDUS
1	Palpable only with Valsalva
2	Easily palpable, not visible
3	Visible

- The veins should decompress supine; **failure to do so, especially on the right, warrants CT or ultrasound for an abdominal or pelvic mass.** The affected testis may be **soft.**
- **Testicular volume is important because it may determine surgery;** it is interpreted against **Tanner stage (Table 55.4).**

METHODS OF MEASURING TESTICULAR VOLUME

Method	Principle	Comment
Prader orchidometer	Chain of 12 solid ellipsoids compared visually with each testis	Reliable
Takahara orchidometer	15 elliptical rings placed over the widest circumference	Reliable
Ultrasound	Length, width and depth with a formula	Most sensitive for left–right differences; volume best calculated as length × width × height × 0.71

TABLE 55.4 TESTICULAR VOLUMES BASED ON TANNER STAGE

Tanner stage	Left testis, cm ³ (mean ± SD) — Daniel et al., 1982	Right testis, cm ³ (mean ± SD) — Daniel et al., 1982	Volume, mL (mean ± SD) — Zachmann et al., 1974
1	4.8 ± 2.8	5.2 ± 3.9	6.0 ± 2.6
2	6.4 ± 3.2	7.1 ± 3.9	6.8 ± 3.6
3	14.6 ± 6.5	14.8 ± 6.1	9.3 ± 3.8
4	19.8 ± 6.2	20.4 ± 6.8	12.6 ± 4.2
5	28.3 ± 8.5	30.2 ± 9.6	16.3 ± 4.6

Tanner stage	Left testis, cm ³ (mean ± SD) — Daniel et al., 1982	Right testis, cm ³ (mean ± SD) — Daniel et al., 1982	Volume, mL (mean ± SD) — Zachmann et al., 1974
6 (Zachmann series only)	—	—	18.9 ± 4.0

- **What the table is for:** it gives the reference against which an affected testis is judged. **Pattern:** volume roughly doubles between Tanner 2 and 3 — the pubertal spurt that makes single measurements unreliable in early puberty — and the **right testis is slightly larger than the left at every stage**. **Trap:** the two series used different methods and give very different absolute values, so **compare a patient only with norms obtained by the same method**, and judge asymmetry side-to-side rather than against a single figure.
- **Ultrasound diagnostic criteria (vein diameter, retrograde flow) are controversial**, more so in adolescents. **Spermatic vein diameter is often normal (≤ 2 mm standing) in grade 1–2 varicoceles** and was a poor criterion when measured supine; **reflux on Valsalva occurs in about two-thirds of grade 2–3 varicoceles**; standing flow velocity correlates with grade and sperm motility; and a **peak retrograde flow on Valsalva above 38 cm/s strongly correlates with testicular volume asymmetry**.

Associated Pathological Processes

- There is abundant evidence that varicocele can impair **testicular growth, spermatogenesis and fertility potential**. The injury is presumed to be **thermal (raised scrotal temperature)**, but the pathogenesis is poorly understood.

Testicular Hypotrophy

- Without semen analysis in most adolescents, **testicular hypotrophy is used as the surrogate of pathology and marker of surgical success**. Reduced volume is long associated with varicocele and rises with grade in some studies but not others.
- **Ipsilateral volume loss may improve or resolve after repair**, taken to mean that hypotrophy reflects real injury. The effect may be **bilateral**: the **right testis is smaller in boys with grade 3 left varicoceles and grows after left repair**, and bilaterally small testes are associated with lower sperm density.
- **"Significant hypotrophy" is variably defined** — a 10%, 15% or 20%, or 2–3 mL, difference — and its reported frequency varies with measurement method, referral pattern and formula. **Pubertal growth can be rapid and asynchronous (age 10–19, by Tanner stage)**, so **serial measurements over years define hypotrophy better than a single one**, especially in early puberty.
- **"Catch-up" growth — normalisation of left relative to right size — follows repair in a third to four-fifths of boys**, but the inference that this reflects improved function is challenged: **the differential may improve spontaneously without treatment**, while in others it **worsens**. Catch-up does **not** depend on lymphatic or arterial preservation; **repair enlarges small but not normal testes**. **Postoperative left hypertrophy** (a left testis over 10% larger) is attributed to **non-lymphatic-sparing repair or a rebound in spermatogenesis**.
- **A significant left–right discrepancy remains the main indication for repair when semen analysis is not obtained** — but **many boys without varicocele have a smaller left testis**, and the link between discrepancy and function is unproven; **randomised prospective studies are needed**.

Testicular Histology

- In adults histology ranges from **normal** (even with severe hypospermatogenesis) to **degenerative tubular changes, altered Leydig-cell number and vascular proliferation**. Adolescent data are scant; abnormal tubular maturation and Leydig-cell numbers at biopsy **did not predict adult sperm quality**.

Hormonal Function

- **Hormonal profiles are inconclusive.** Basal LH and FSH are not consistently different from controls; stimulated levels do not improve after surgery; an **exaggerated GnRH response** in some patients does not correlate with hypotrophy; **inhibin B does not consistently correlate** with size or semen.
- More recently, **total motile sperm count (TMC; normal above 9 million per ejaculate) rose with total testicular volume and inhibin B and fell with FSH**, and cut-offs were proposed as **surrogates for TMC** in boys deferring semen analysis; a **higher reflux grade was associated with hypotrophy and higher FSH**.

Semen Quality

- **Semen quality is the most useful indicator of fertility potential**, despite collection and interpretation difficulties. Quality improves with testicular growth and maturation, so **analysis is best obtained once testicular growth stabilises**. Ejaculation begins at a bone age of about 13 years and normospermia is reached by a bone age of about 17; LH and FSH fall as normospermia is attained, though some Tanner 5 boys remain asthenozoospermic.
- **No age- or Tanner-based norms exist**, so studies use **adult World Health Organization criteria**, usually on at least two samples; a **single analysis may suffice because TMC is reproducible**.
- In **randomised studies** of treated versus untreated boys, semen quality was similar except for a **postoperative rise in sperm concentration** (a presumed rebound). Total testicular volume predicts TMC; in one series the differential did not, but in another **a differential above 20% doubled, and a total volume under 30 mL quadrupled, the odds of a low TMC**.
- **Grade and postoperative catch-up growth do not reliably predict ultimate semen quality; poorer sperm quality may be confined to a subset of boys with varicocele**.

Intratesticular Varicocele

- In 1–2% of adolescents the varicocele has an **intratesticular component on CDUS: anechoic structures over 2 mm near the mediastinum testis with venous flow that increases on Valsalva**. Asymmetry worsened without treatment and responded to repair, so **surgery is indicated when an intratesticular varicocele coexists with testicular asymmetry**.

Concept — *Every adolescent varicocele marker is a surrogate: volume, grade, hormones and histology are used because semen is unavailable, and none reliably predicts adult semen quality — which is why observation, not repair, is the default.*

Treatment

- **Observation is the approach for most adolescents until a surgical indication appears.**
- **Indications for surgery: significant left (> 20%) or bilateral testicular hypotrophy, pain, or abnormal semen analysis** — the last **most reliable at Tanner 5 and/or age 18 or older**. Pain is a rare indication (2–10%) and is **relieved in most (68–88%)**.
- **Practice varies widely** in indications, approach, follow-up and use of semen analysis; **few urologists order semen analysis routinely**, and many are uncomfortable raising it.

Surgical Repair of Varicocele

- Options: **inguinal or subinguinal, retroperitoneal (open or laparoscopic), or venographic**. The decision turns on (1) **whether to spare the testicular artery and/or lymphatics**, and (2) **the resulting rates of recurrence and hydrocele**.
- **Long-term data are lacking**: follow-up is often too short to capture recurrence or **hydrocele, which may take 2 years or more to appear**; many hydroceles resolve, but **up to half need scrotal repair or aspiration**. Artery-sparing and artery-ligating laparoscopic techniques improve semen quality equally.

TABLE 55.5 RESULTS OF ADOLESCENT VARICOCELE REPAIR

Procedure	Recurrence or persistence	Hydrocele	Testicular atrophy
Open suprainguinal (Palomo)	2–4%	0–30% (meta-analysis 10%)	—
Laparoscopic, non-lymphatic- or non-artery-sparing	0–9%	11–32% (meta-analysis 7%)	—
Laparoscopic, artery- and/or lymphatic-sparing	1–7%	0–4%	—
Microscopic subinguinal	0–10%	0–6%	Rare
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy	6–35%	Occasional	Rare

Meta-analysis values from Barroso et al., 2009.

- **What the table tests:** hydrocele is the price of not sparing lymphatics (mass ligation — Palomo, non-sparing laparoscopy), and recurrence is the price of less invasive or less magnified techniques (non-microscopic inguinal, sclerotherapy). **Microscopic subinguinal repair combines low recurrence with low hydrocele, but is the only approach with (rare) atrophy.** **Trap:** the meta-analysis hydrocele rate for non-sparing laparoscopy lies outside its own range — cite the range as the tested value.

Subinguinal or Inguinal Microsurgical Varicocelectomy

- The **subinguinal microscopic approach** makes **artery and lymphatic sparing easier**, with **high success and low hydrocele risk**, but is **time-consuming and needs microsurgical skill**; the technique mirrors the adult procedure (Chapter 70). **Loupe magnification gives inferior results to the operating microscope.** It is associated, rarely, with **testicular atrophy (necrosis)**, not reported after suprainguinal repair.

Retroperitoneal and Laparoscopic Varicocelectomy

- **Ligation of the internal spermatic vessels above the internal ring**, open or laparoscopic, is simple and highly successful but has **higher hydrocele rates**. **Mass ligation (Palomo) is highly successful and efficient.**
- **Lymphatic- and/or artery-sparing techniques**, though harder, **reduce hydrocele but may lower success**, and **do not improve catch-up growth**; artery sparing was associated with better semen quality in one series, but **no randomised data define its effect on testicular function**.
- **Atrophy risk:** in theory after **artery ligation in a boy with prior inguinal surgery** (though not seen after laparoscopic Palomo in a small series), and after a **non-artery-sparing repair followed by surgery that compromises the vasal artery or vein** (hernia repair, vasectomy).
- **Open Palomo sequence:** **muscle-splitting incision medial to the anterior superior iliac spine** → mobilise the peritoneum medially → expose and isolate the spermatic vessel bundle → **clip or ligate, ± divide**. A **microscopic lymphatic-sparing** modification through this incision had no recurrences and was **faster and more reliable than a subinguinal approach**.
- **Laparoscopic sequence:** **three ports** (working ports in variable positions) → open the peritoneum **above the internal ring** → mobilise the vascular bundle → identify the **artery (laparoscopic Doppler or papaverine)** and **lymphatics (clear tubular channels, aided by magnification)** — **vital dyes** injected paratesticularly help but risk **injection into the testis** → ligate with sutures or clips and divide (harmonic scalpel or sealing device) → **ligate the veins accompanying the artery to prevent recurrence**; dilated refluxing **deferential veins** found on CDUS may also be ligated.
- **Genitofemoral nerve injury**, commoner with **electrocautery**, causes **anterior thigh paraesthesia that resolves slowly over weeks to months**.

Sclerotherapy or Embolotherapy

- Adolescent experience is limited, mostly for **recurrent or persistent varicocele**. Its particular benefit is to **identify and classify venous collaterals** as routes of outflow and reflux; **aberrant vasculature is found in about a quarter**. Agents: **3% sodium tetradecyl sulfate or polidocanol**, $\pm$ intravascular coils or balloons.

RETROGRADE VERSUS ANTEGRADE SCLEROTHERAPY

Feature	Retrograde	Antegrade
Access	Transfemoral venous puncture under local anaesthesia $\pm$ sedation	Small subinguinal or upper scrotal incision ; cannulate a vein ($\pm$ fluoroscopic confirmation of flow into the internal spermatic vein)
Technique	Venographic embolisation/sclerosis	Inject sclerosant, then ligate the vein
Results	Inferior to surgery; technical failure 5–22%; recurrence 6–35% ; follow-up rarely beyond 1–2 years	Recurrence 4–12% — better than retrograde
Complications	Self-limiting pain, epididymo-orchitis, phlebitis, scrotal swelling; partial atrophy and hydrocele rare ; radiation exposure	Similar

KEY POINTS: VARICOCELE

- **Prevalence is similar in adolescents and adults — about 15%.**
- **Tall, thin boys more often have varicoceles**, and a **genetic predisposition** is implicated.
- Varicoceles are commonly found on examination and **least commonly because of pain**.
- **Testicular volume is best measured by ultrasound.**
- **Hypotrophy of 15–20%** may accompany an ipsilateral varicocele; it commonly resolves after repair **but can resolve spontaneously**.
- Poorer sperm quality may be limited to a **subset**; **grade and catch-up growth do not reliably predict semen quality**.
- **Success rates are similar for retroperitoneal (open or laparoscopic) and subinguinal approaches**; **lymphatic-sparing techniques have the lowest hydrocele rates**, and most hydroceles that form need surgery.

Concept — Every varicocele technique trades hydrocele against recurrence: mass ligation buys certainty at the cost of the lymphatics, sparing techniques protect the lymphatics at the risk of missed veins, and microscopic magnification is how the subinguinal approach achieves both.

EPIDIDYMAL AND VASAL ANOMALIES

- **Congenital anomalies of the accessory testicular structures often accompany disorders of testicular descent and genital development**, but may be isolated or part of syndromes that spare the genitalia.
- **Epididymal cysts** are simple cysts, palpable or found **incidentally on ultrasound**; they are common and **rise in frequency with age** (about a third of scanned boys over 15). They are **clinically indistinguishable from spermatoceles**, except that spermatoceles are postpubertal and contain sperm. Their cause is unknown but may be **hormonal** — they are linked to **diethylstilbestrol (DES) exposure**. They differ on ultrasound and pathology from the **multicystic solid epididymal cystadenomas of von Hippel–Lindau disease**. Many resolve spontaneously; surgery is rarely needed.
- **Congenital absence of the vas deferens (CAVD)** is usually due to **mutations of the cystic fibrosis transmembrane conductance regulator (CFTR) gene** that are milder than those of cystic fibrosis. It may

be associated with **renal agenesis or ectopia** and partial or complete **agenesis of the epididymis and seminal vesicles**, and causes **infertility**.

BILATERAL VERSUS UNILATERAL CONGENITAL ABSENCE OF THE VAS

Feature	Bilateral (CBAVD)	Unilateral (CUAVD)
Dominant association	CFTR mutations commoner	Renal anomalies commoner
Contralateral vas	—	Normal or obstructed
Consequence	Infertility (obstructive)	Infertility if contralateral obstructed

- **Persistent mesonephric duct:** abnormal vasal development can leave a **persistent mesonephric duct with ipsilateral renal and seminal vesicle agenesis and vasoureteral fusion; reflux into the fused Wolffian remnants causes UTI and epididymo-orchitis**. Associated anomalies include **imperforate anus, hypospadias, tracheo-oesophageal fistula, tetralogy of Fallot and coarctation of the aorta**.

Concept — *Vasal, seminal vesicle and ureteric anomalies share a mesonephric (Wolffian) origin, which is why absence of the vas keeps company with ipsilateral renal anomalies — and why, when the absence is bilateral, the CFTR gene rather than the kidney is the dominant association.*